

Simulation Based

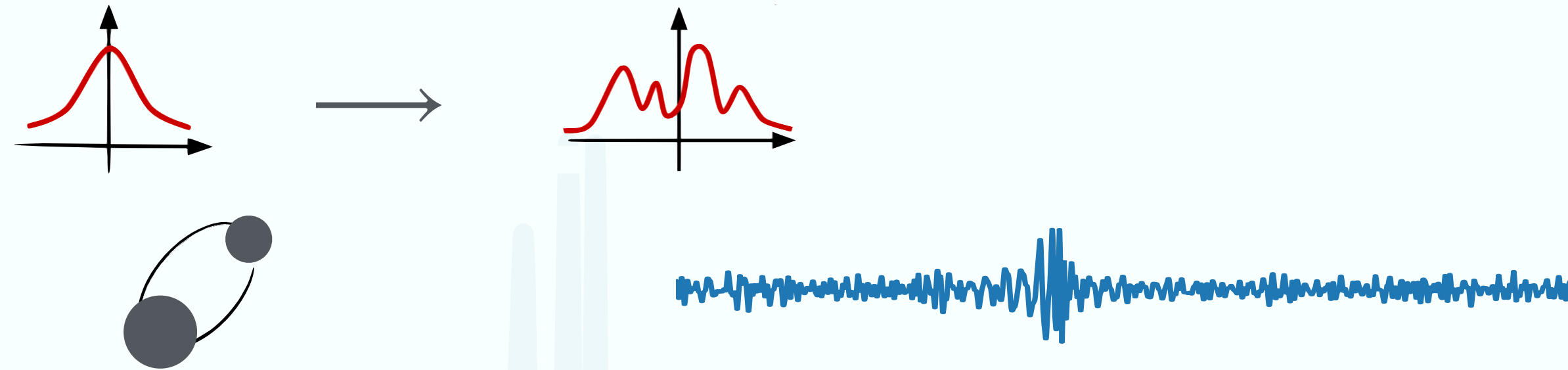
Inference

a.k.a. Likelihood-free Inference

Overview

- Double intro

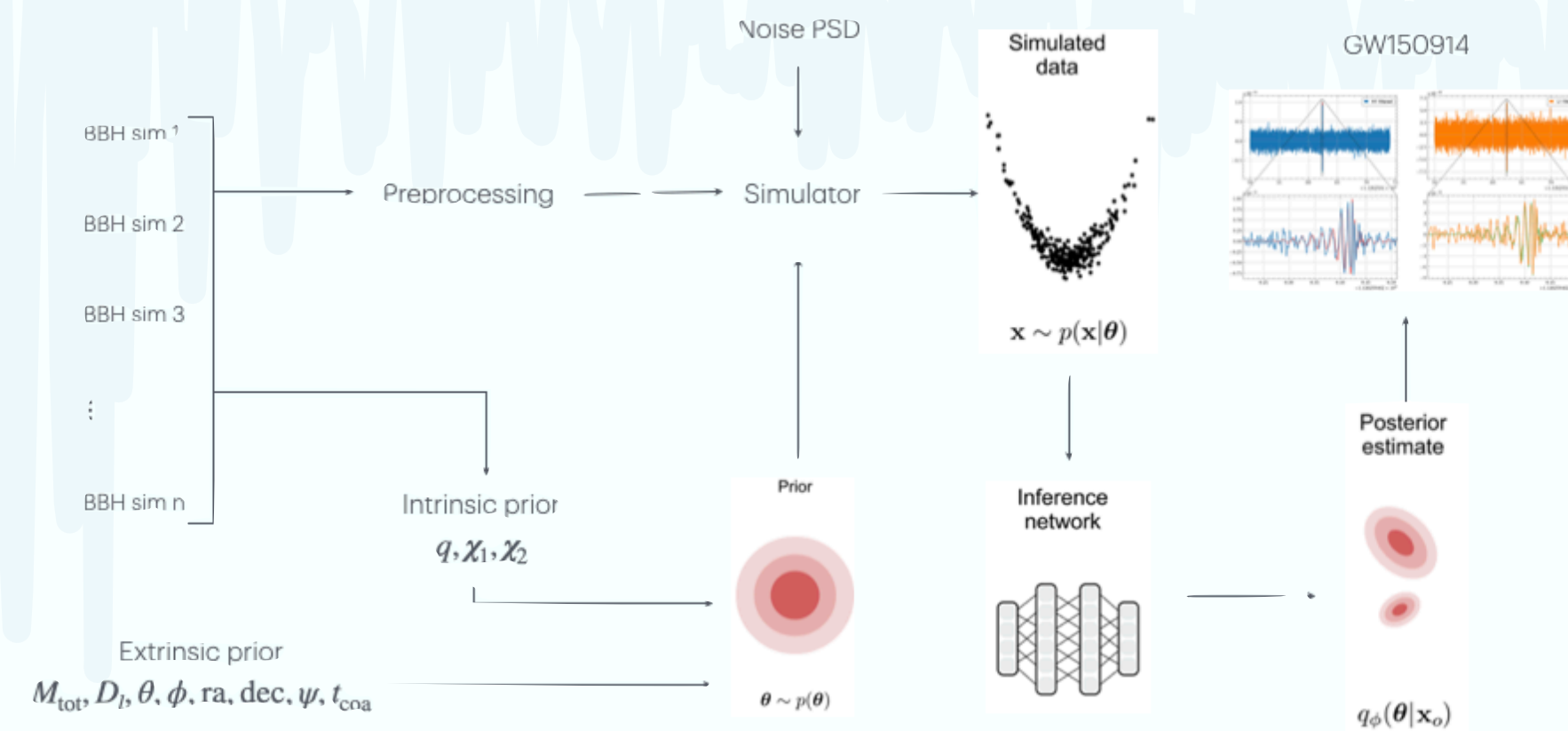
- What is SBI?
- GW astronomy



- Waveform systematics 🤔

- GW without waveform approximant

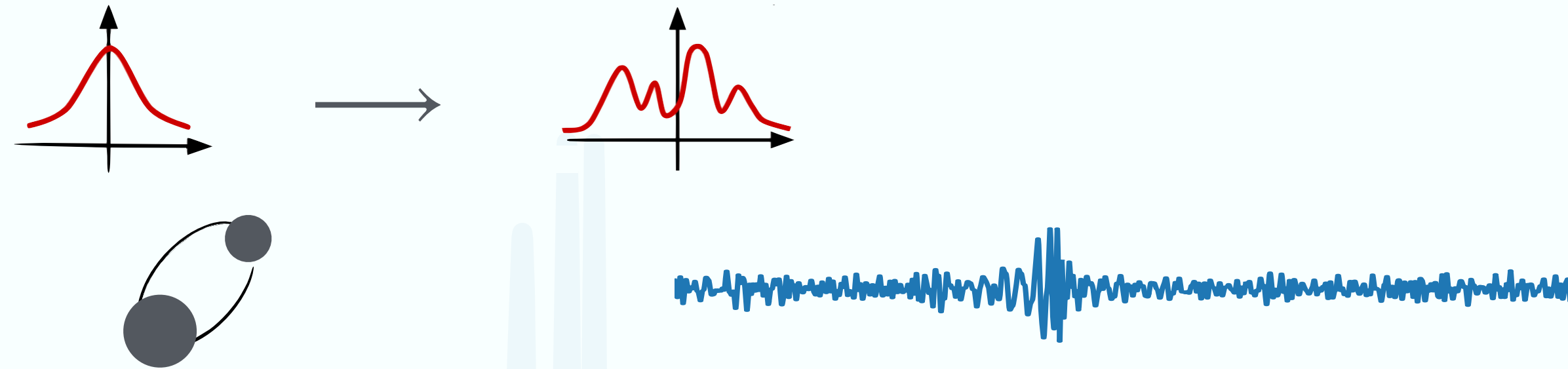
- Setup
- Challenges
- Preliminary results



Overview

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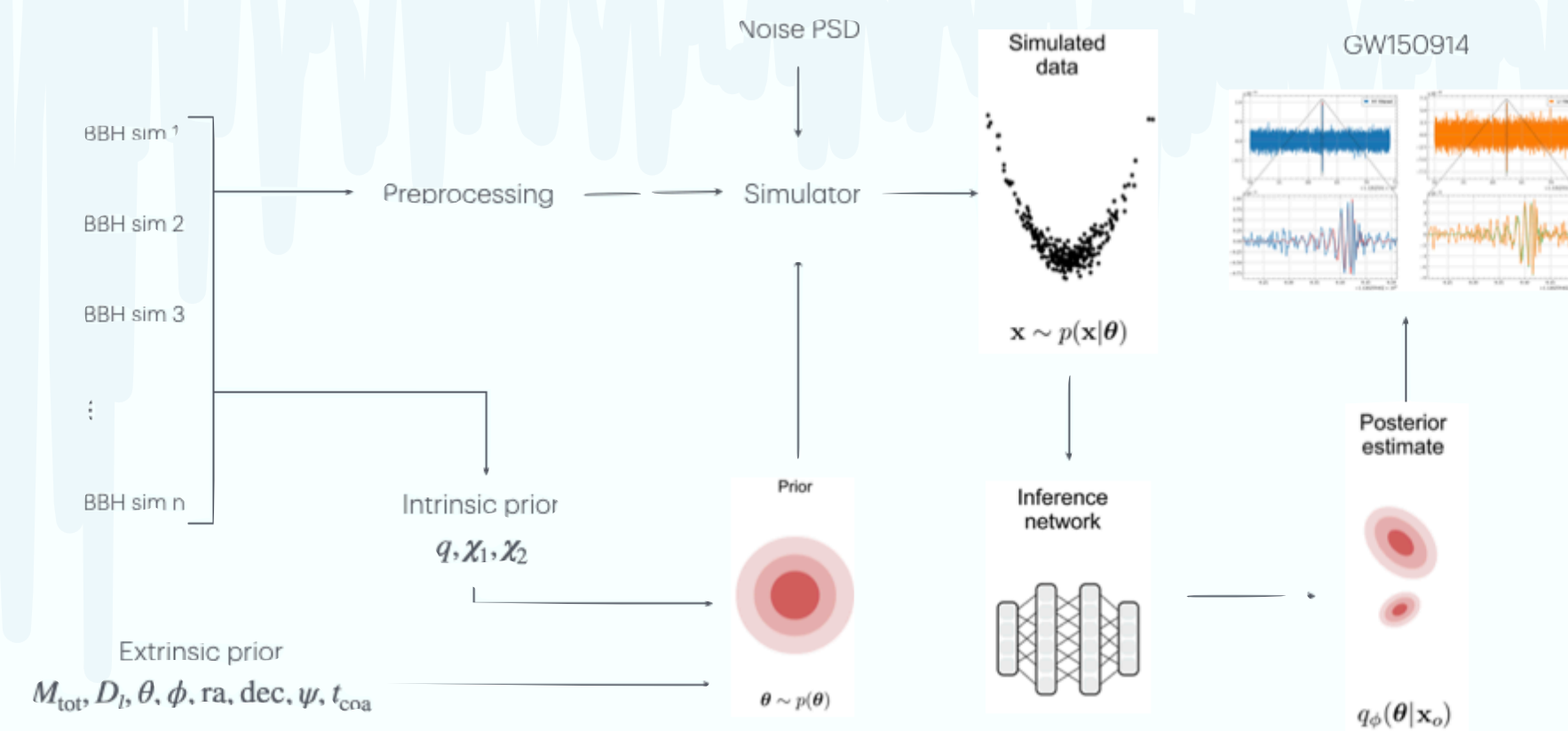


This is not a course on SBI,
it's an ongoing research projects

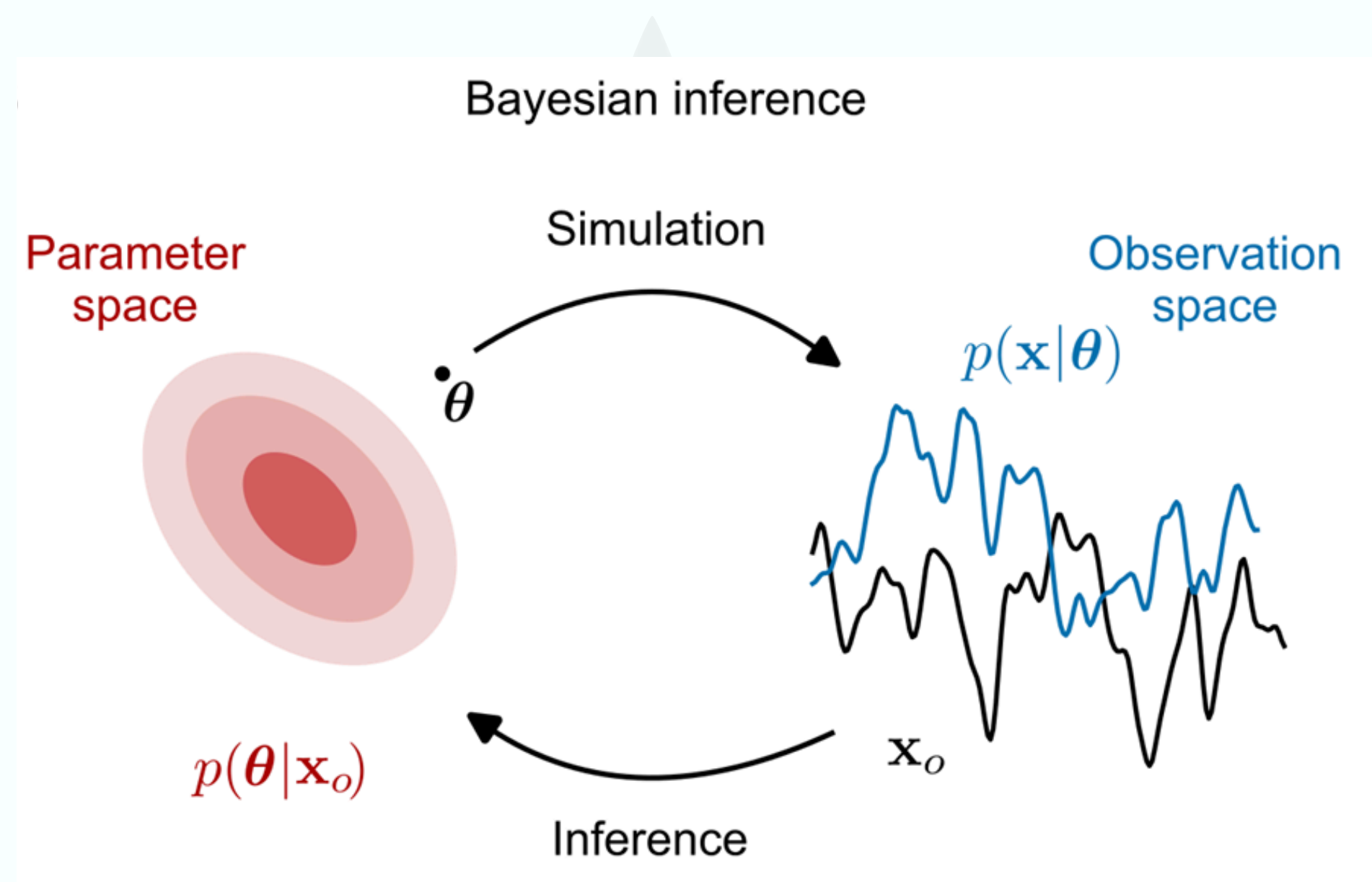
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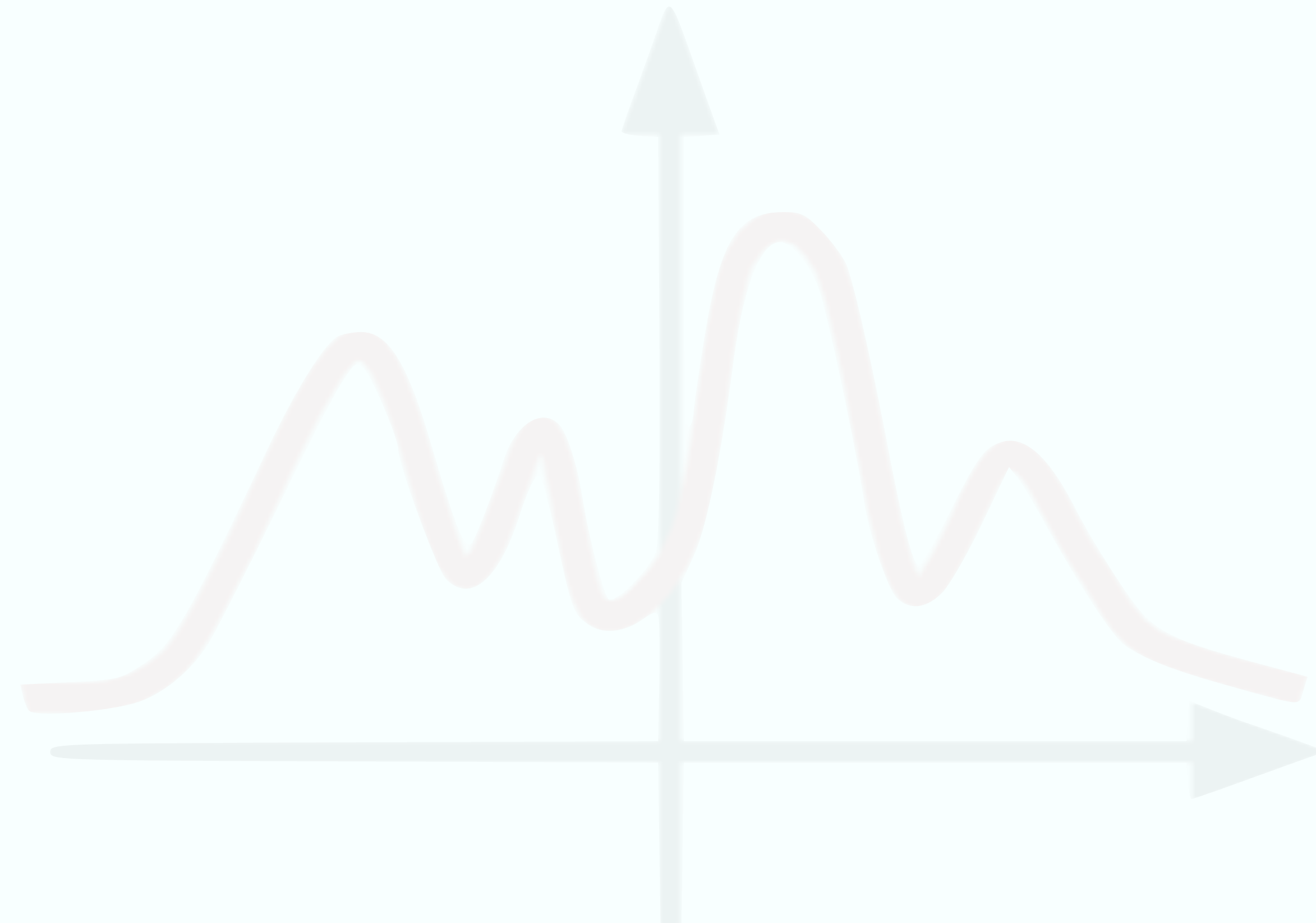
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- Challenges
- Preliminary results



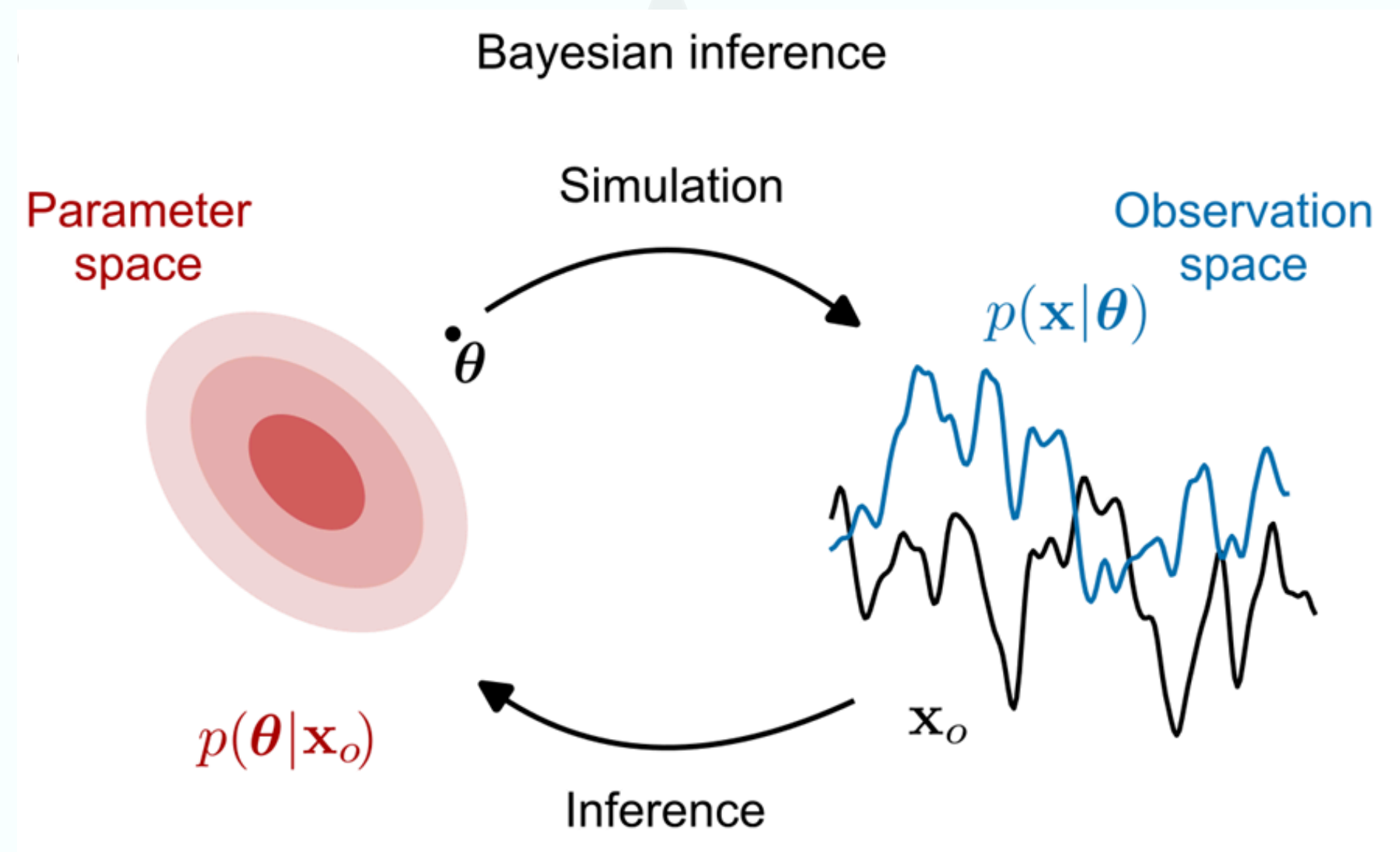
SBI in a nutshell



Deistler+25



SBI in a nutshell



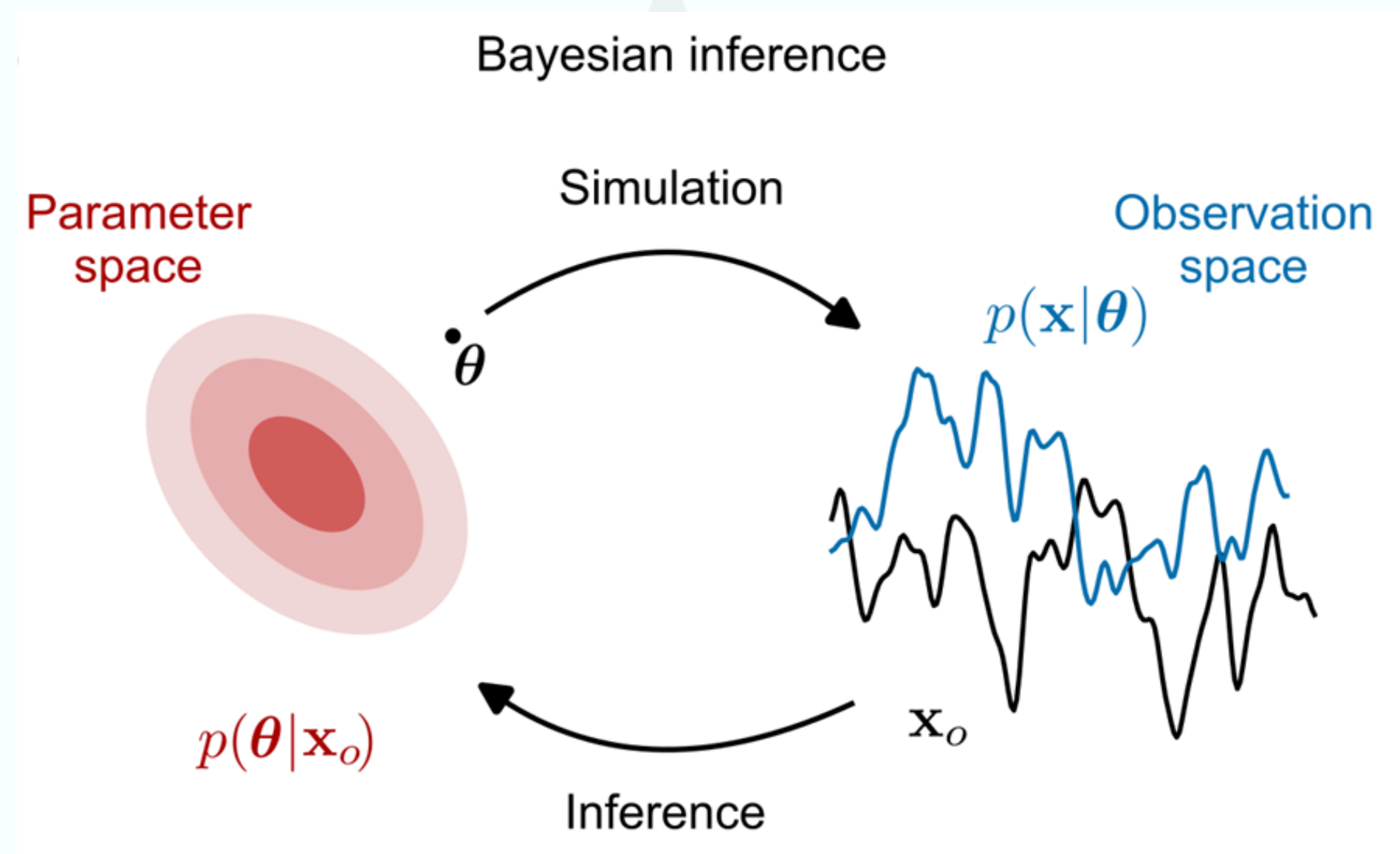
Deistler+25



MCMC



SBI in a nutshell

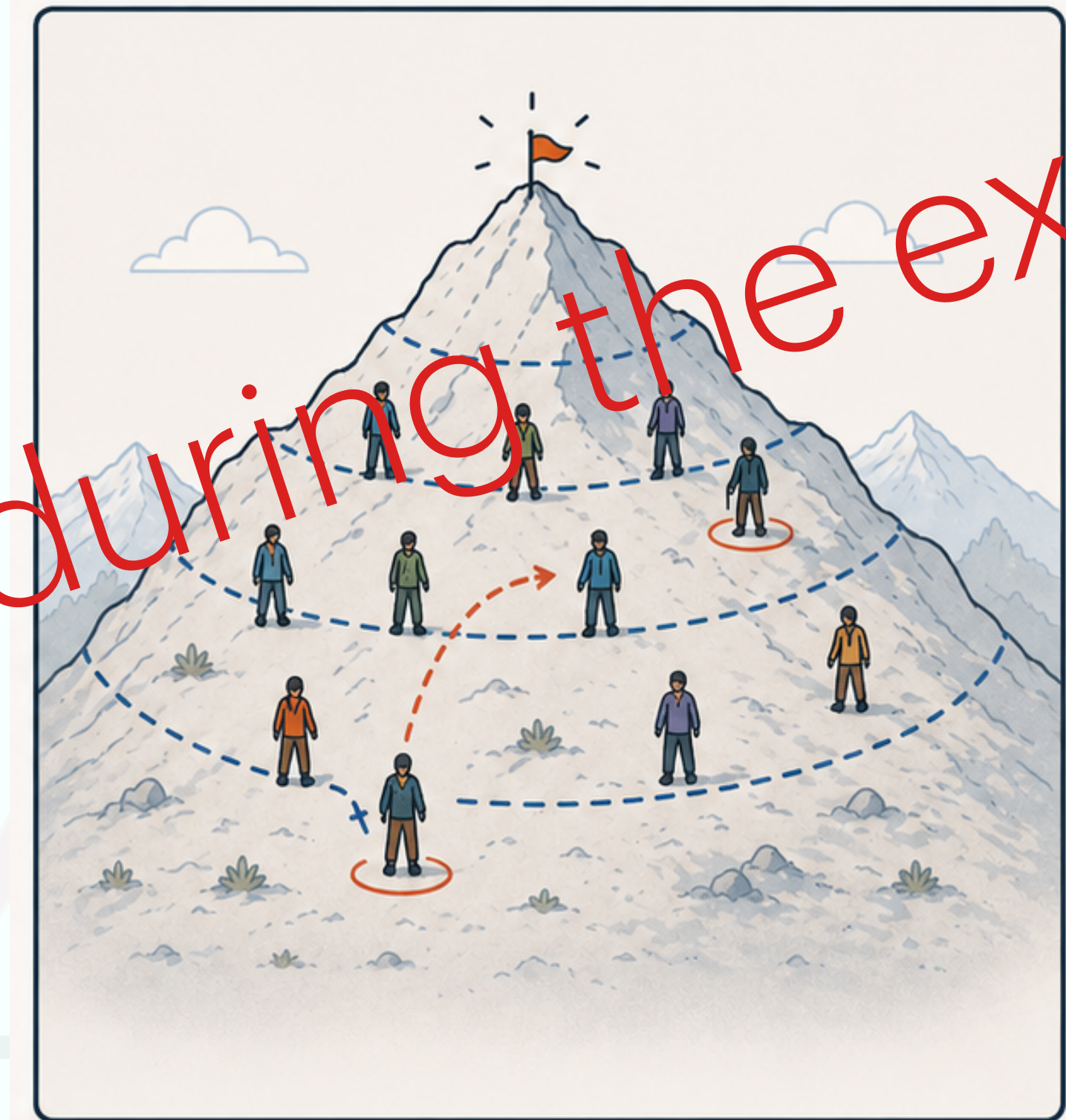


Deistler+25

Nested sampling

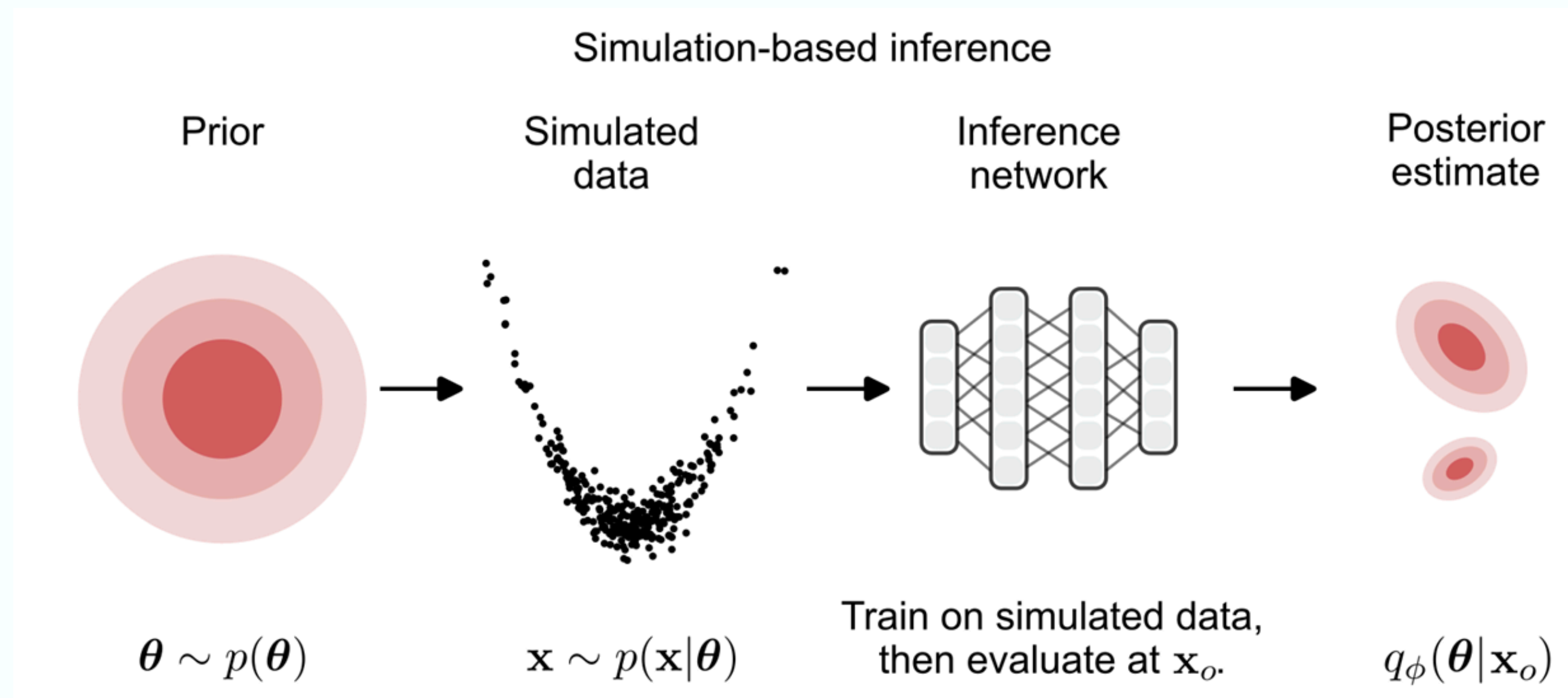


SBI in a nutshell

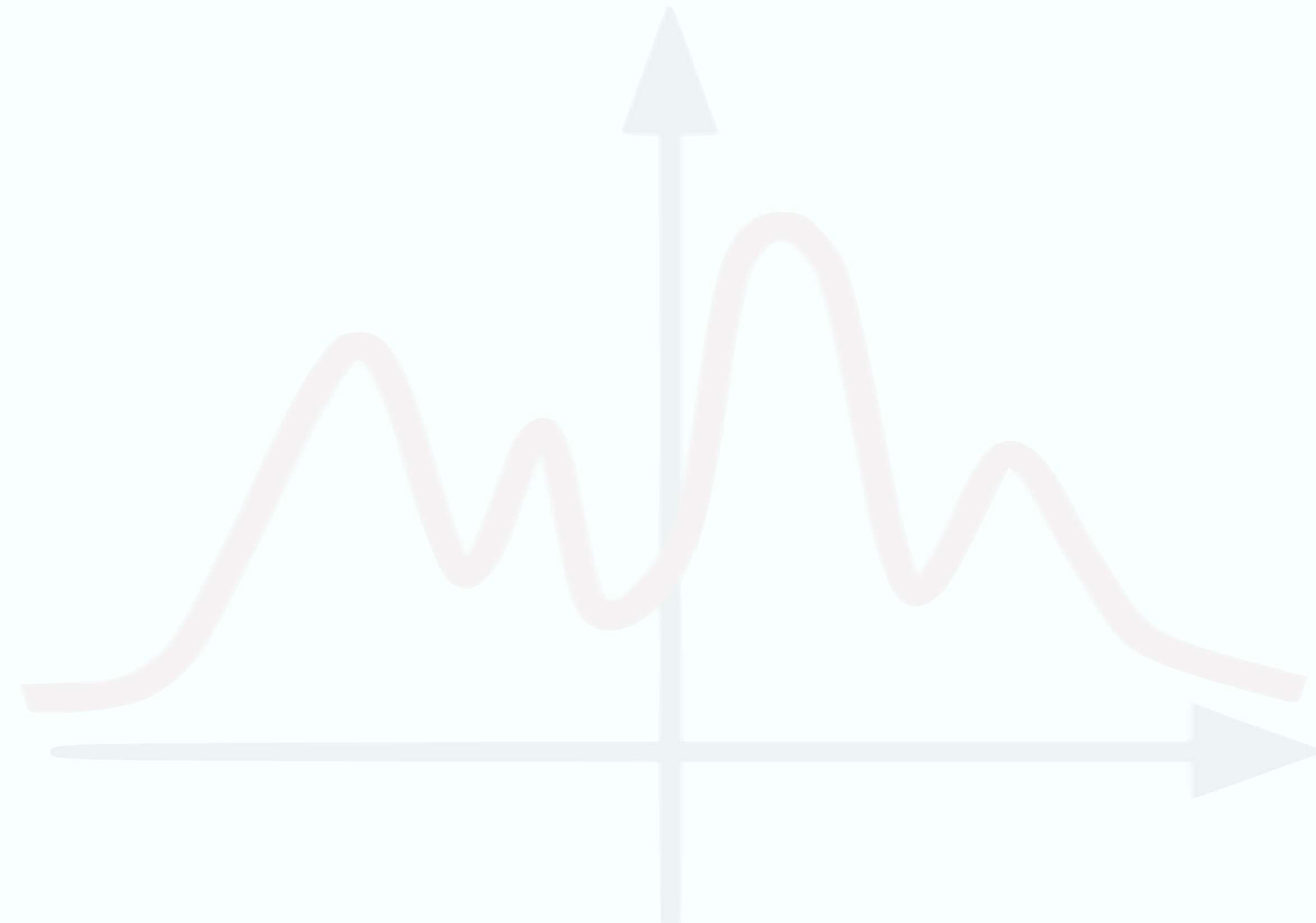


Do NOT use this analogy during the exam!

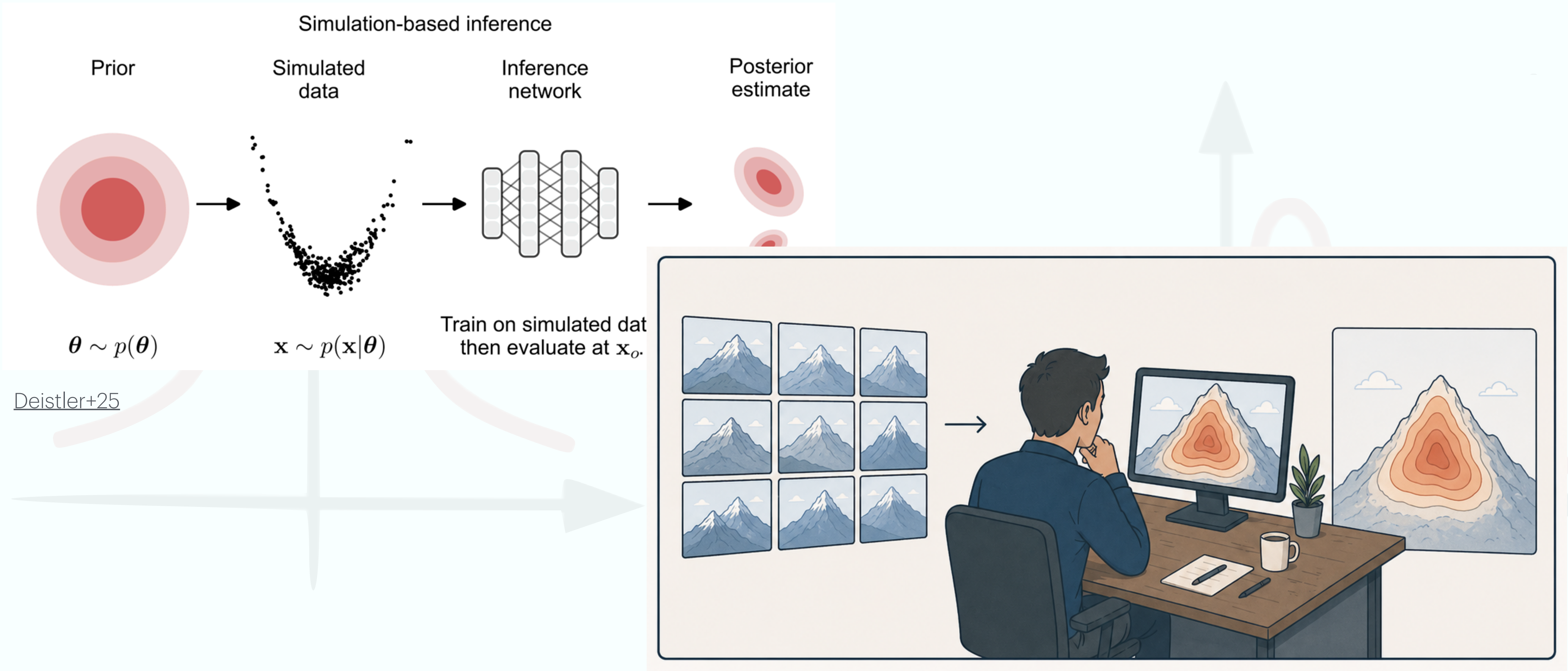
SBI in a nutshell



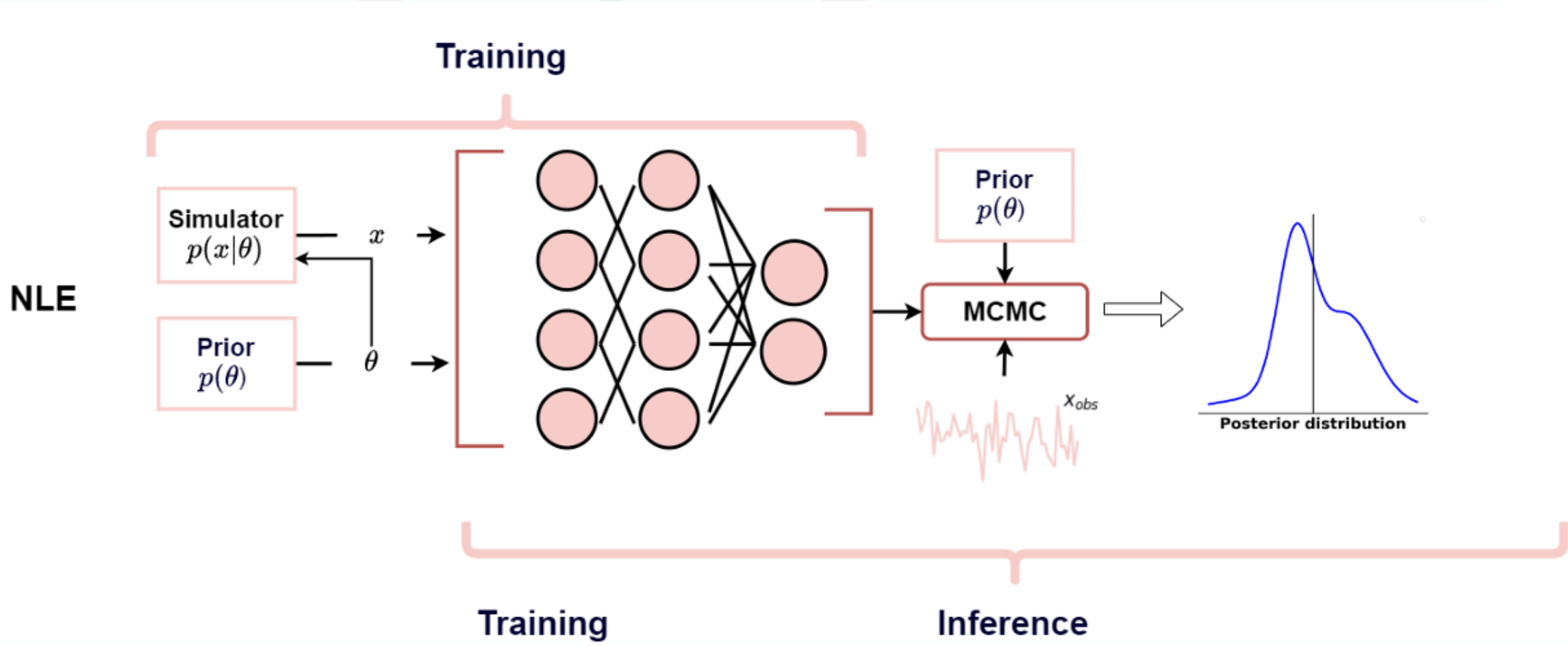
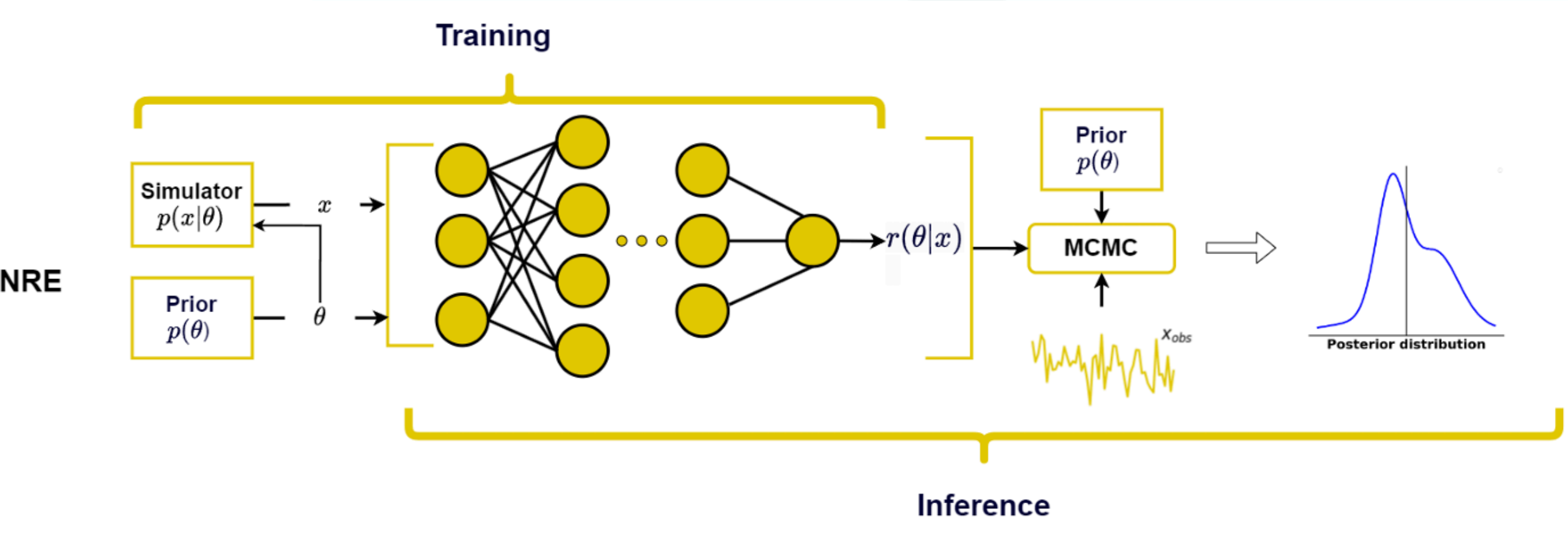
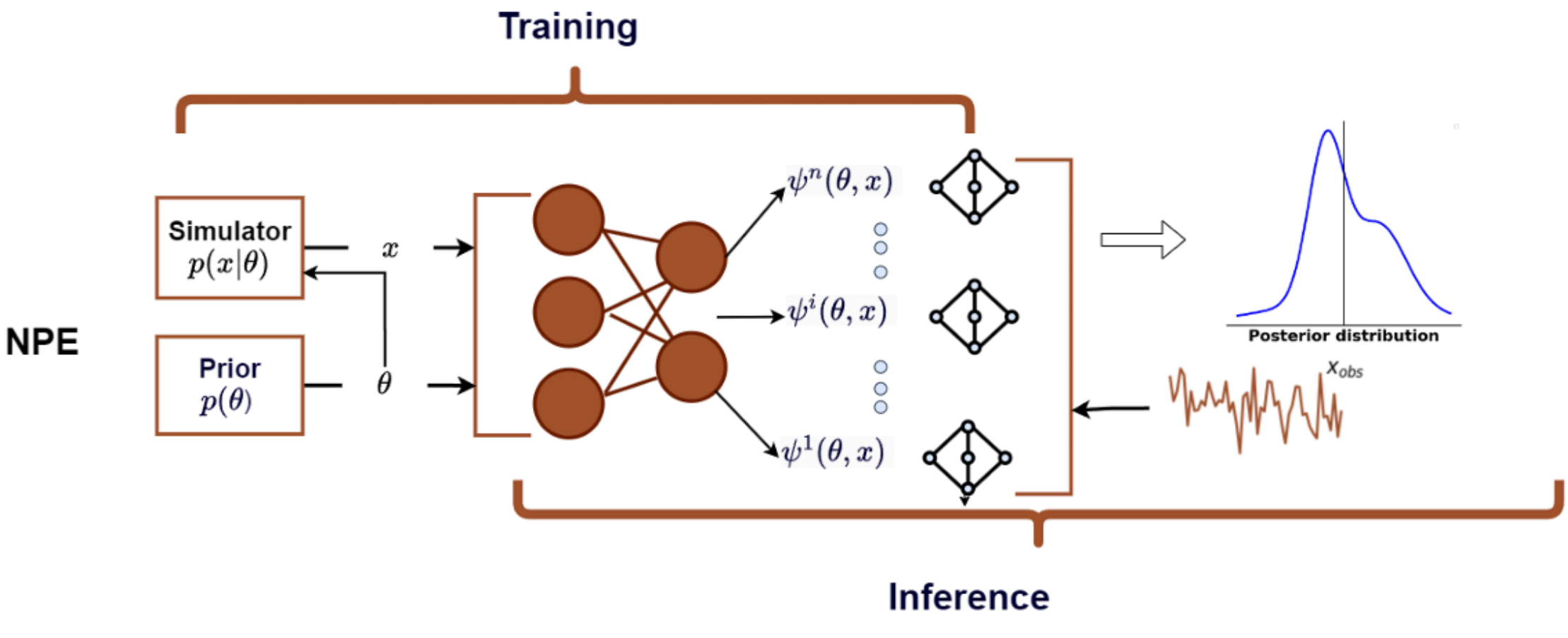
[Deistler+25](#)



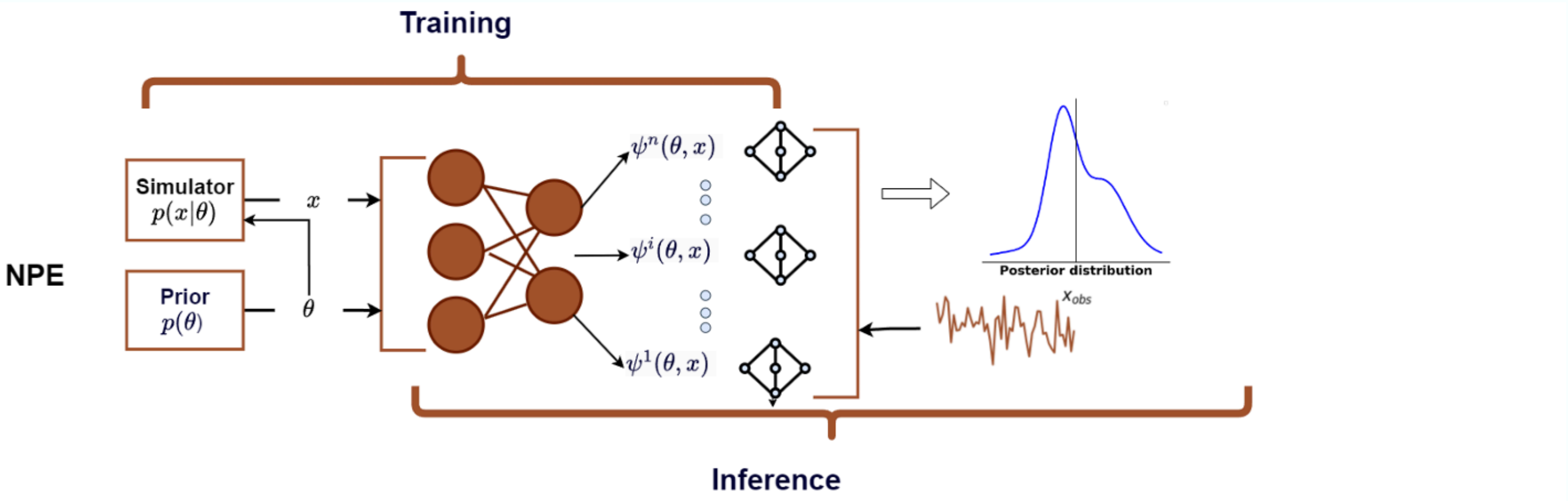
SBI in a nutshell



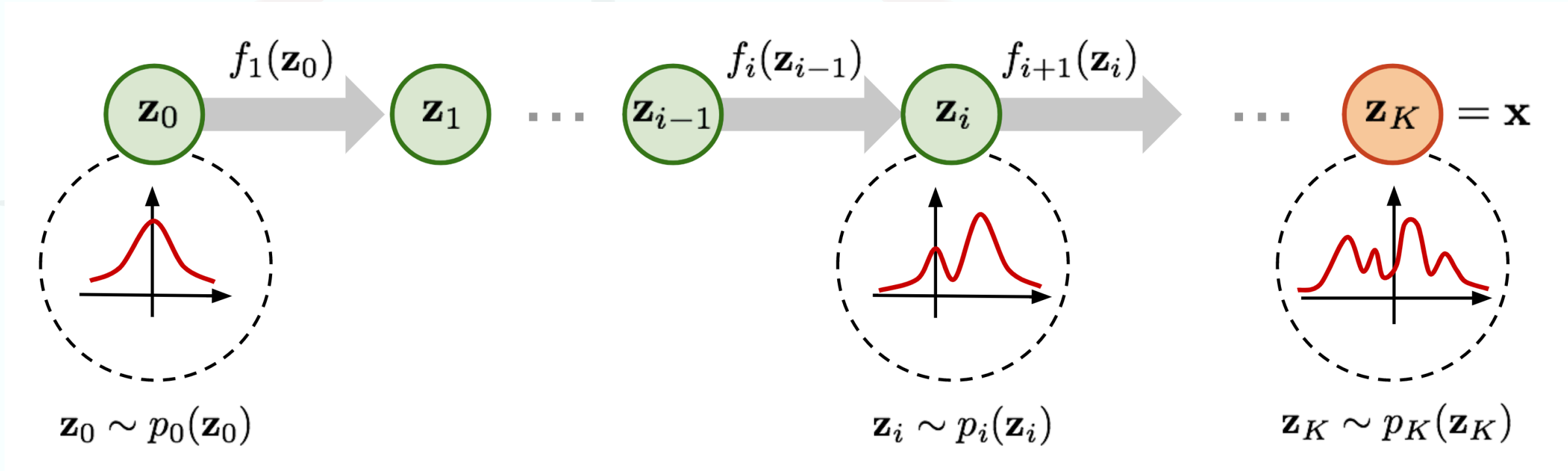
SBI in a nutshell



SBI in a nutshell



[Liang&Wang25](#)



[Lilian Weng](#)

$$q(\theta|x) = q_0(\psi_x^{-1}(\theta)) \left| \det \frac{d\psi_x^{-1}}{d\theta} \right|$$

$$q(\theta|x) \sim p(\theta|x)$$

$$L_{NPE} = - \mathbb{E}_{\theta \sim p(\theta)} \mathbb{E}_{x \sim p(x|\theta)} \log q(\theta|x)$$

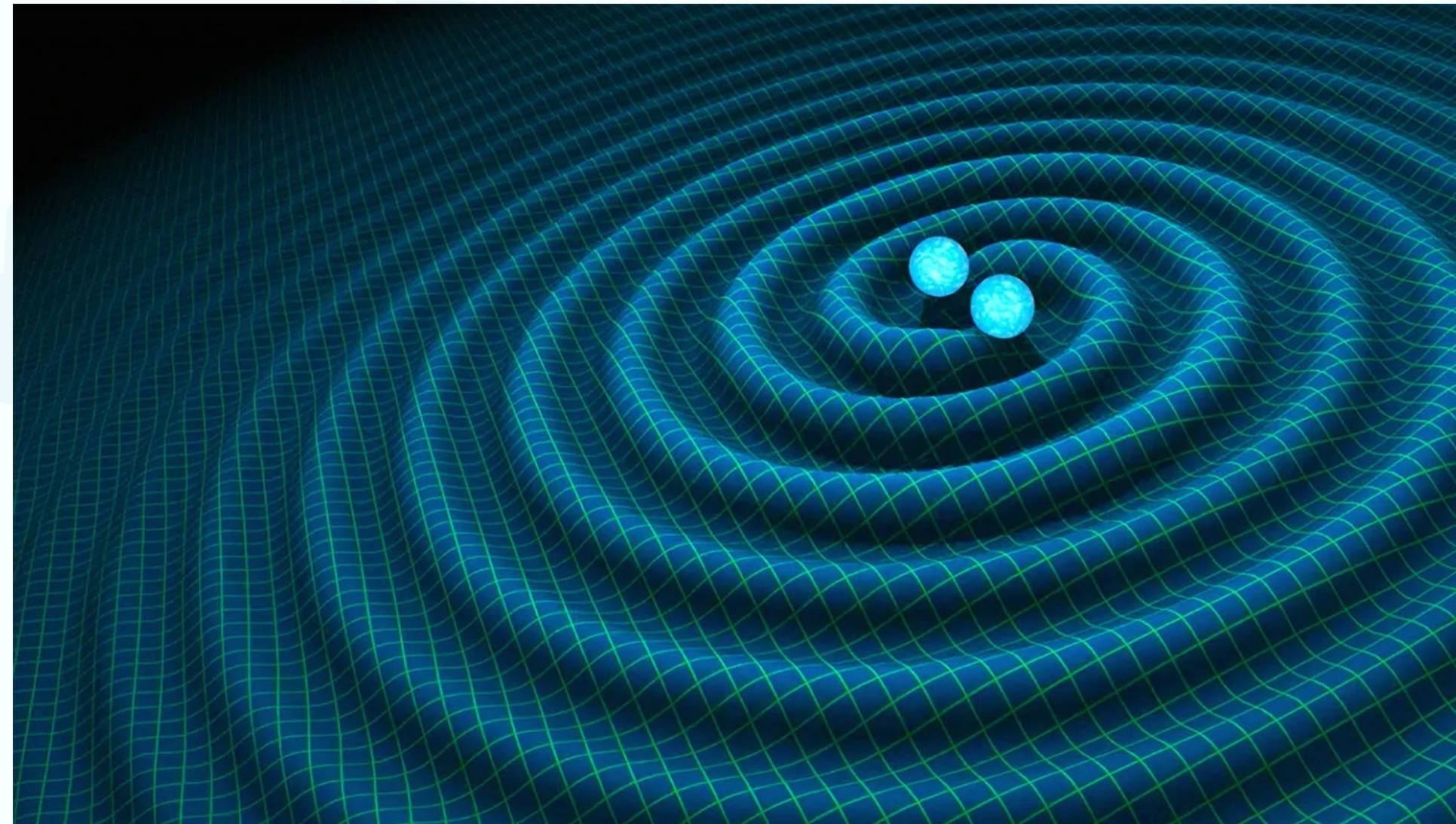
Crash course on GW astronomy

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi T_{\mu\nu} \quad \longrightarrow \quad \square h = 0$$

These slides are neither meant to
substitute nor summarize
the whole picture. Please attend
Relativistic Astrophysics, GW Astronomy
and General Relativity courses

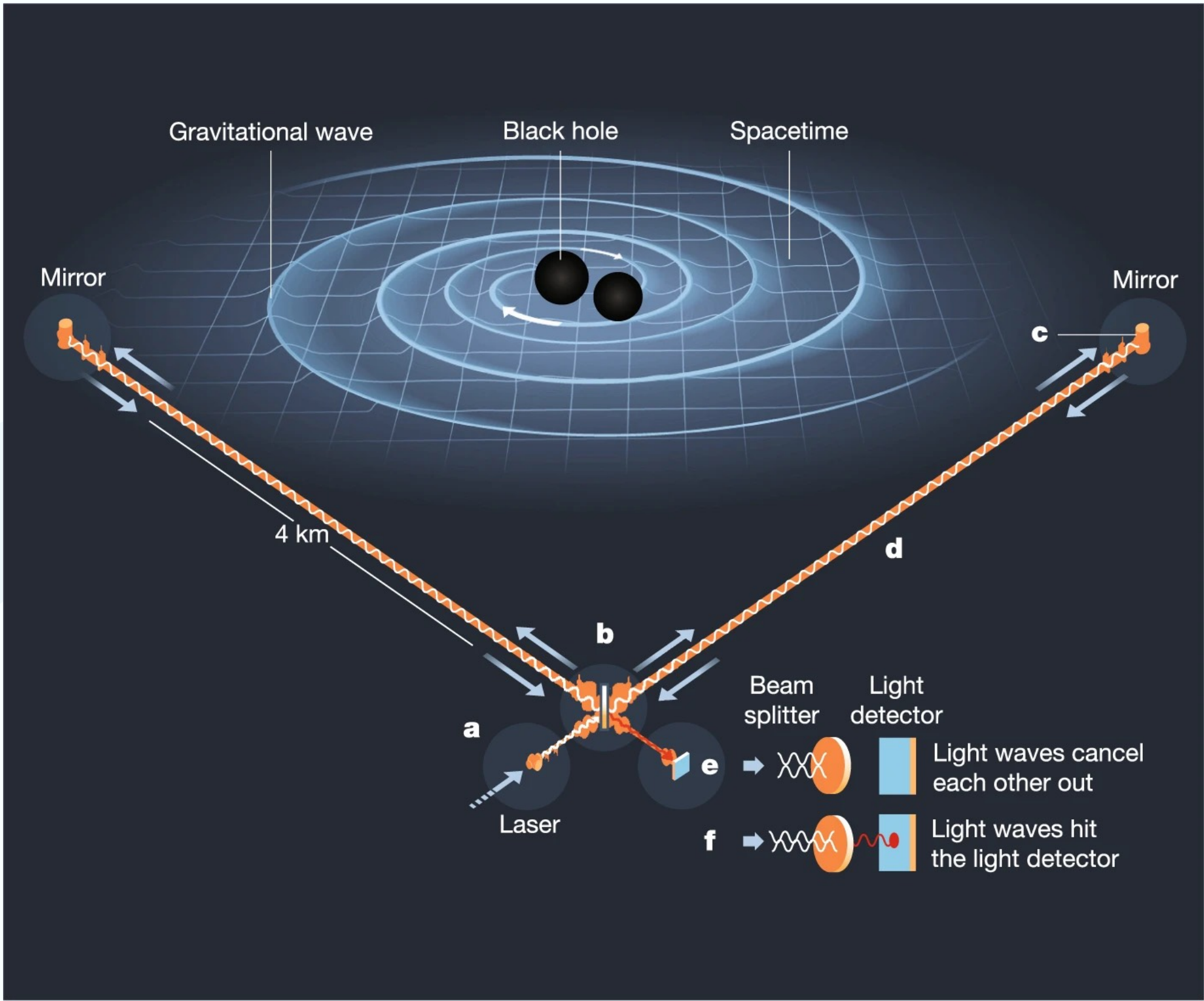
Crash course on GW astronomy

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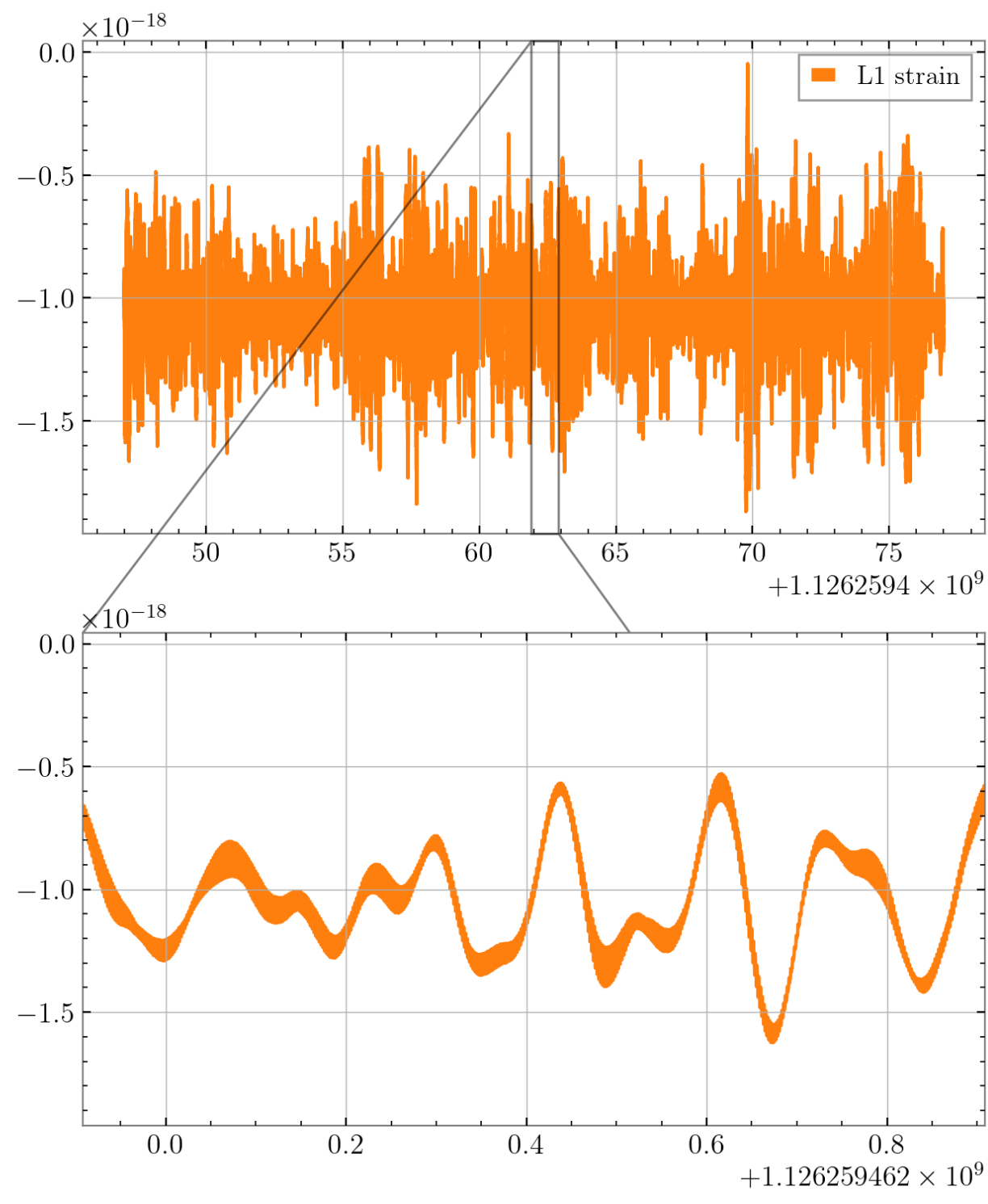
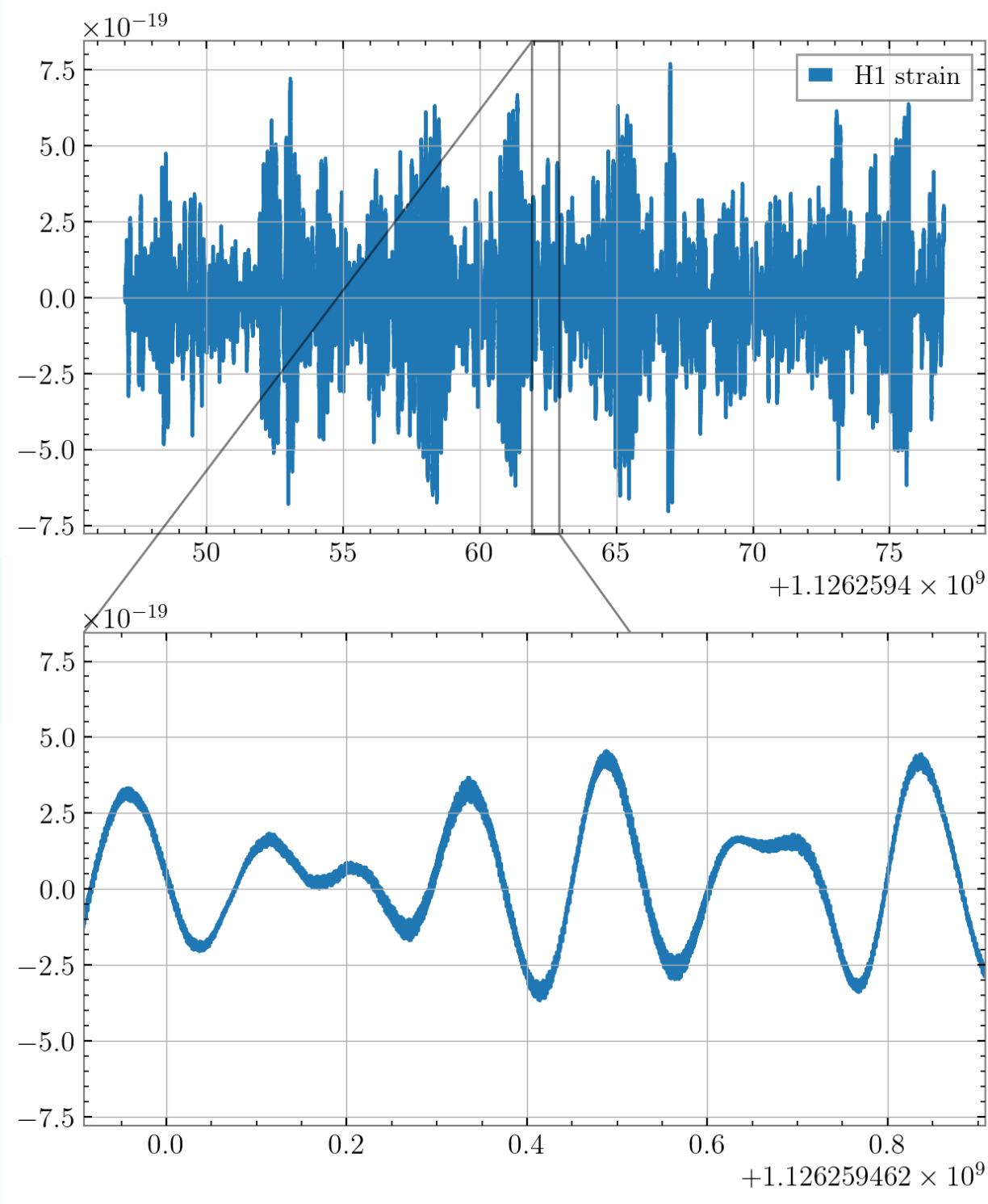


<https://www.einsteintelelescope-emr.eu/>

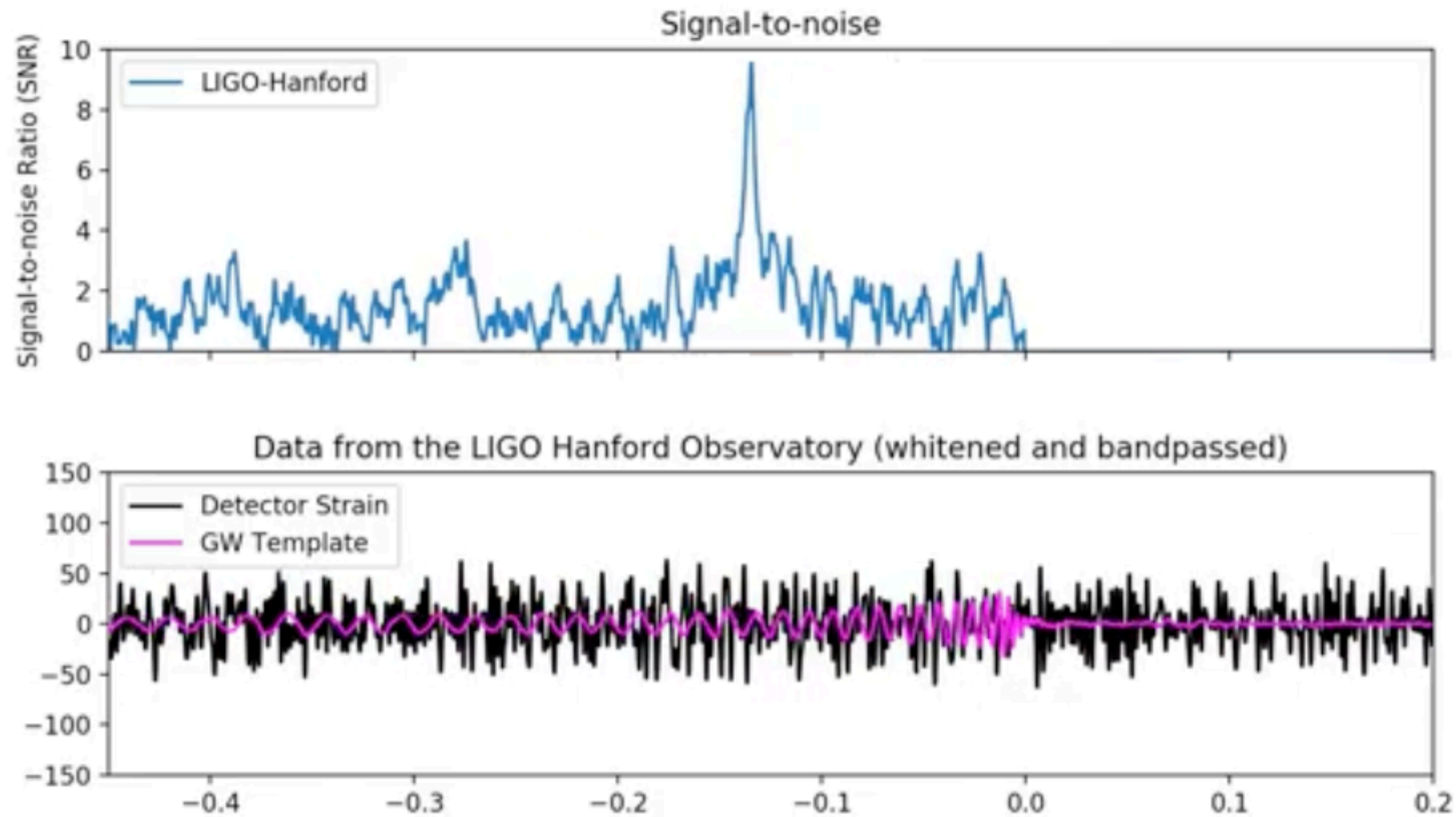
Crash course on GW astronomy



Miller&Yunes19

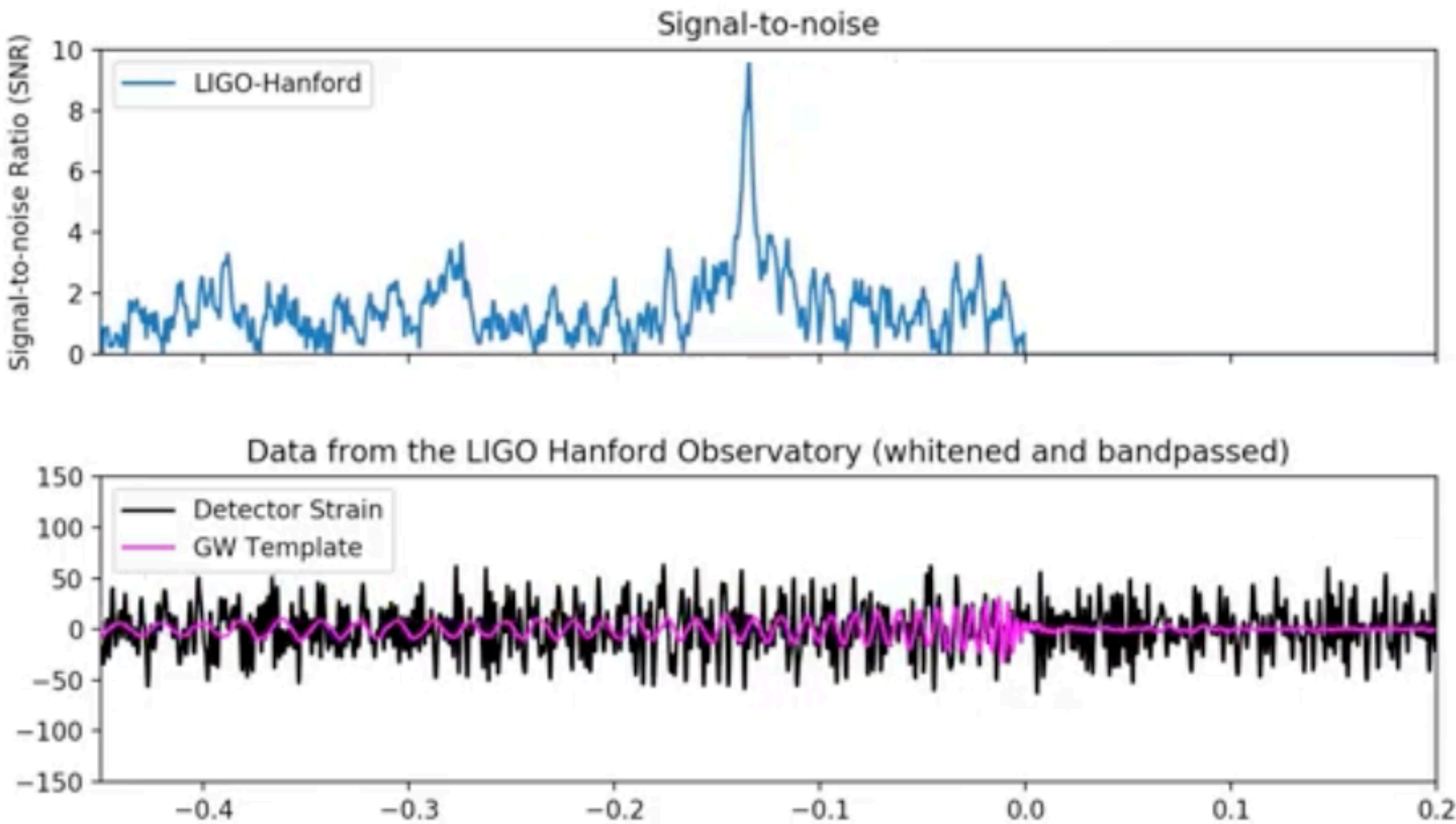


Crash course on GW astronomy

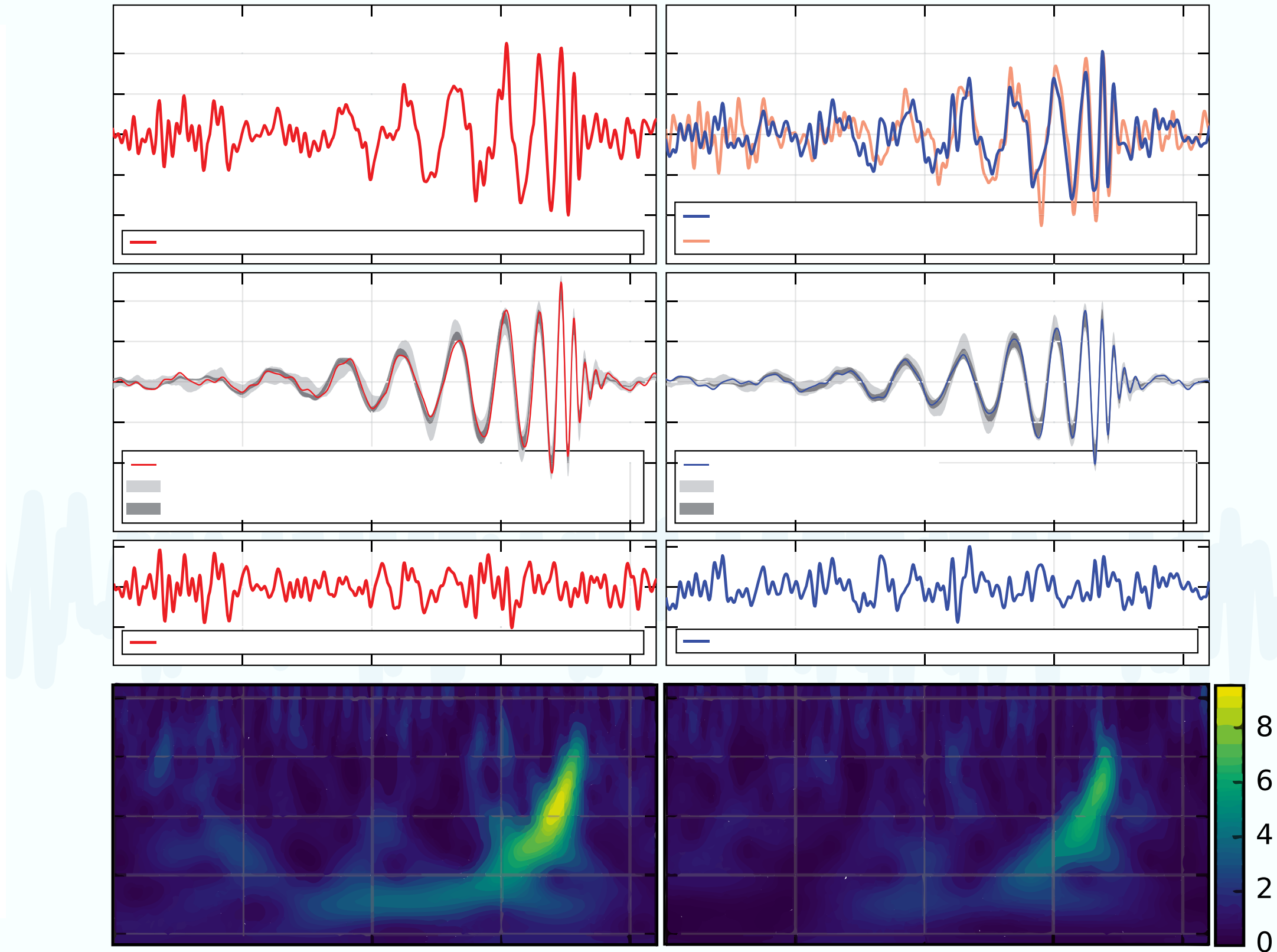


A. Nitz

Crash course on GW astronomy

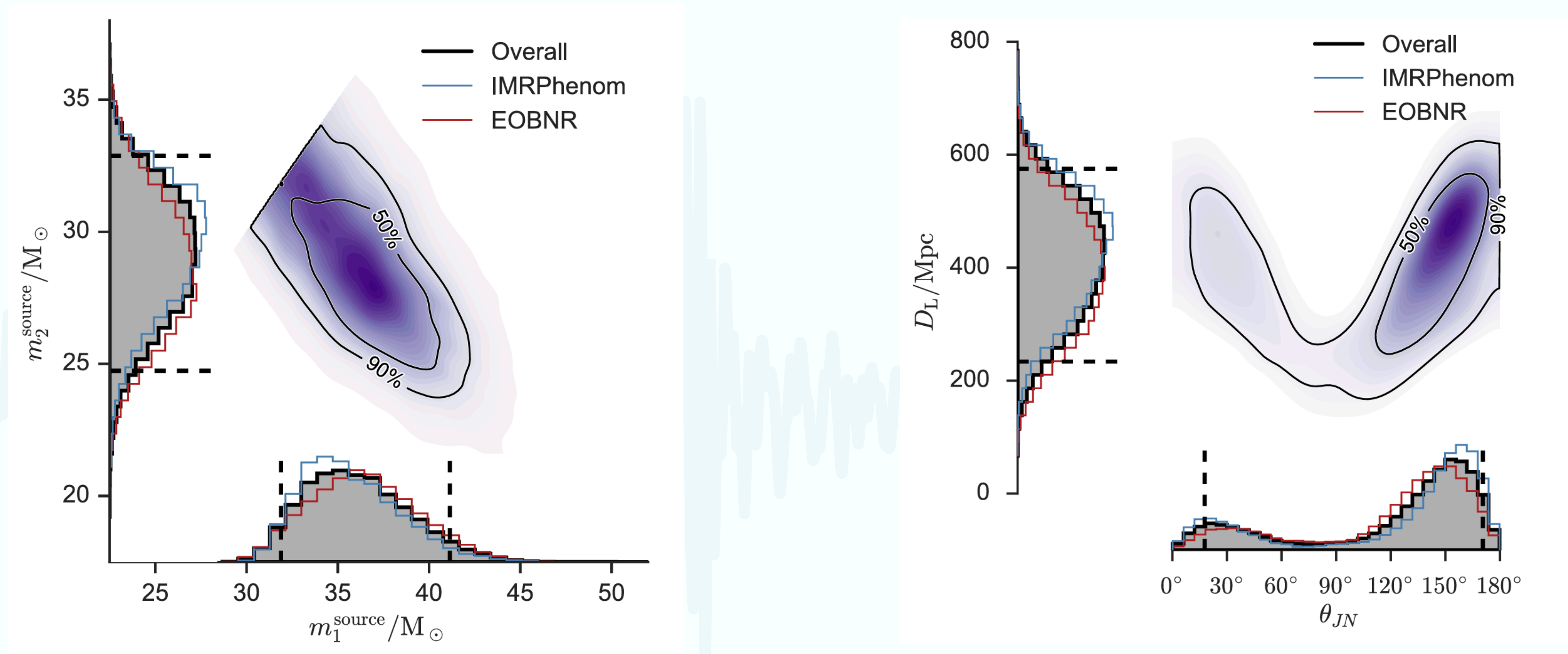


A. Nitz



Abbot+16

Crash course on GW astronomy



Abbot+16

Approximants of approximants

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi T_{\mu\nu}$$



Approximations



Waveform
approximants



Phenomen

EOB

NRSur

PE
&
Science

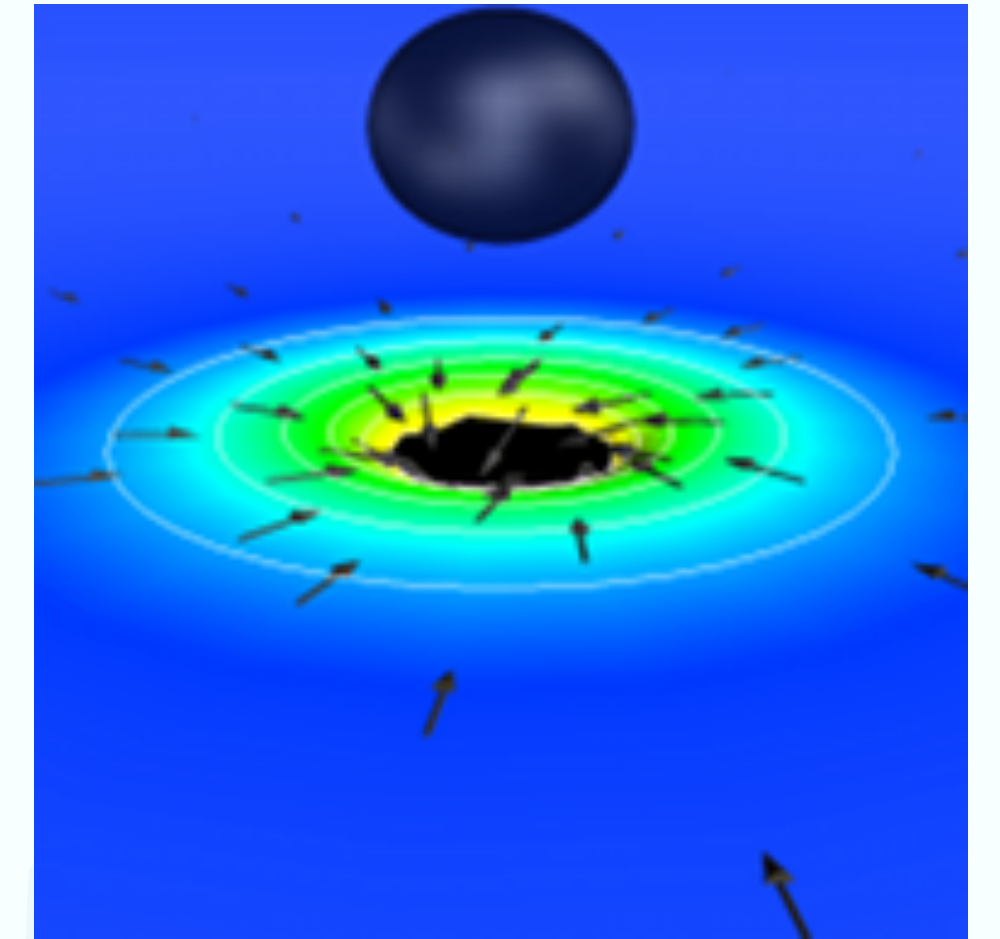
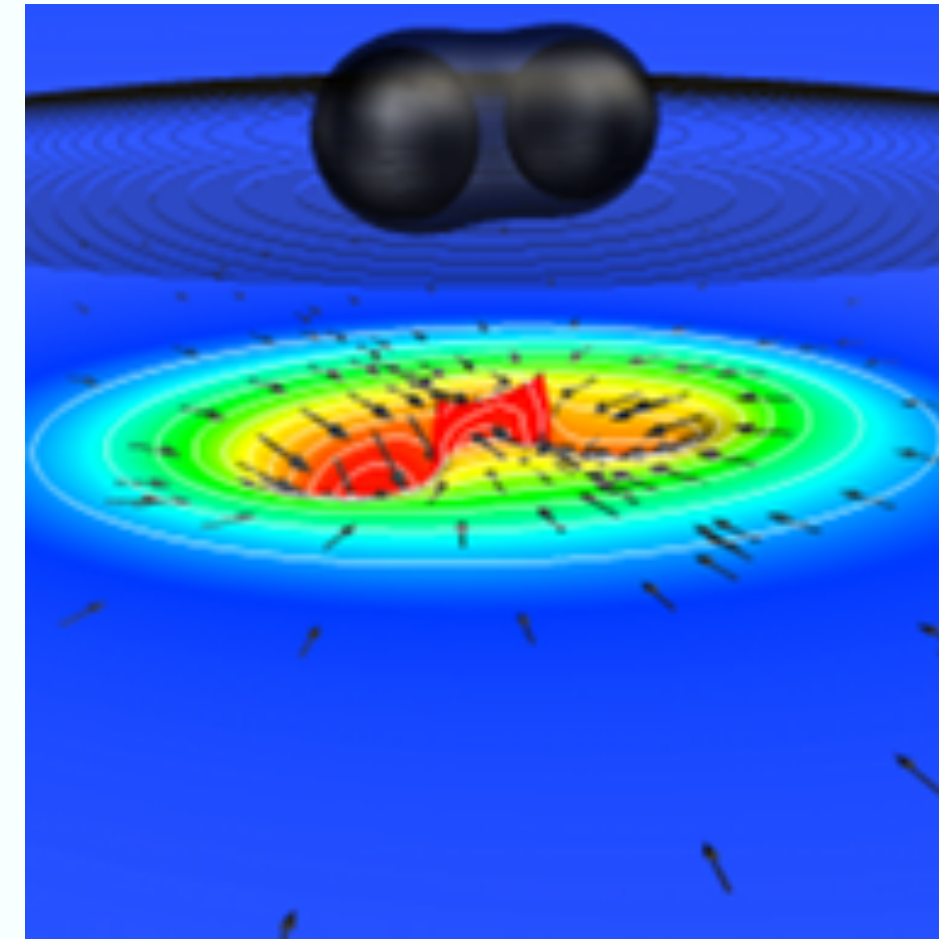
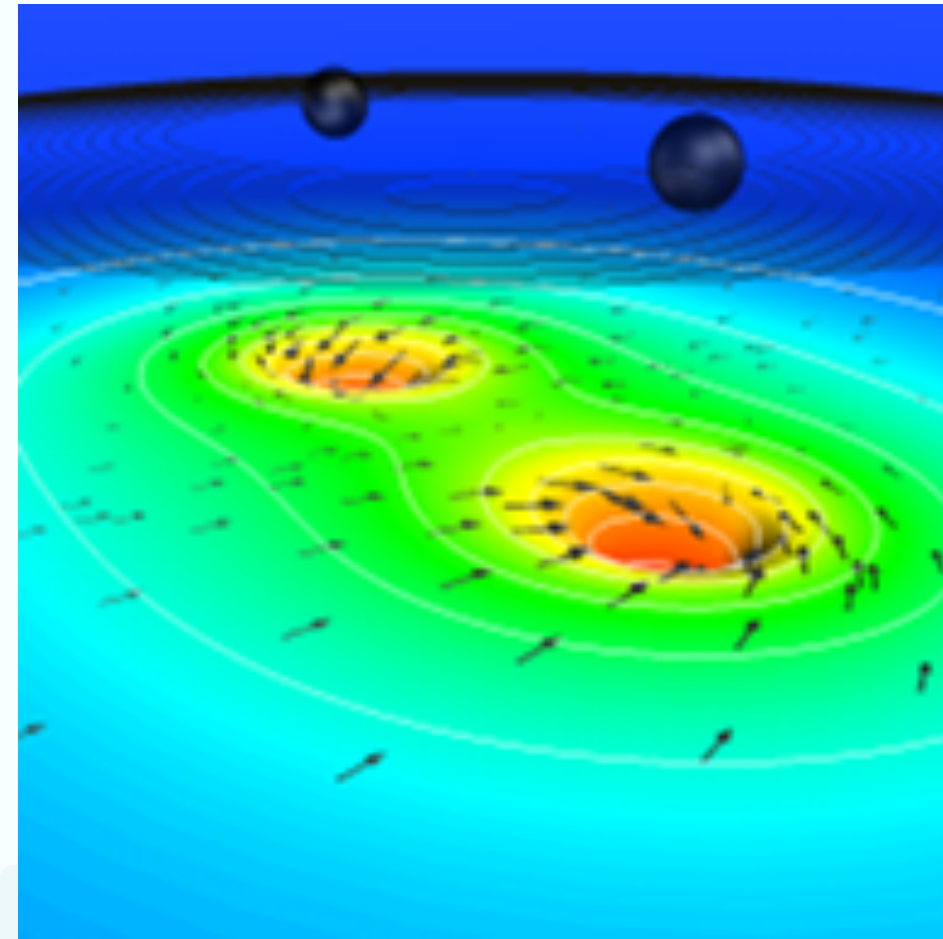
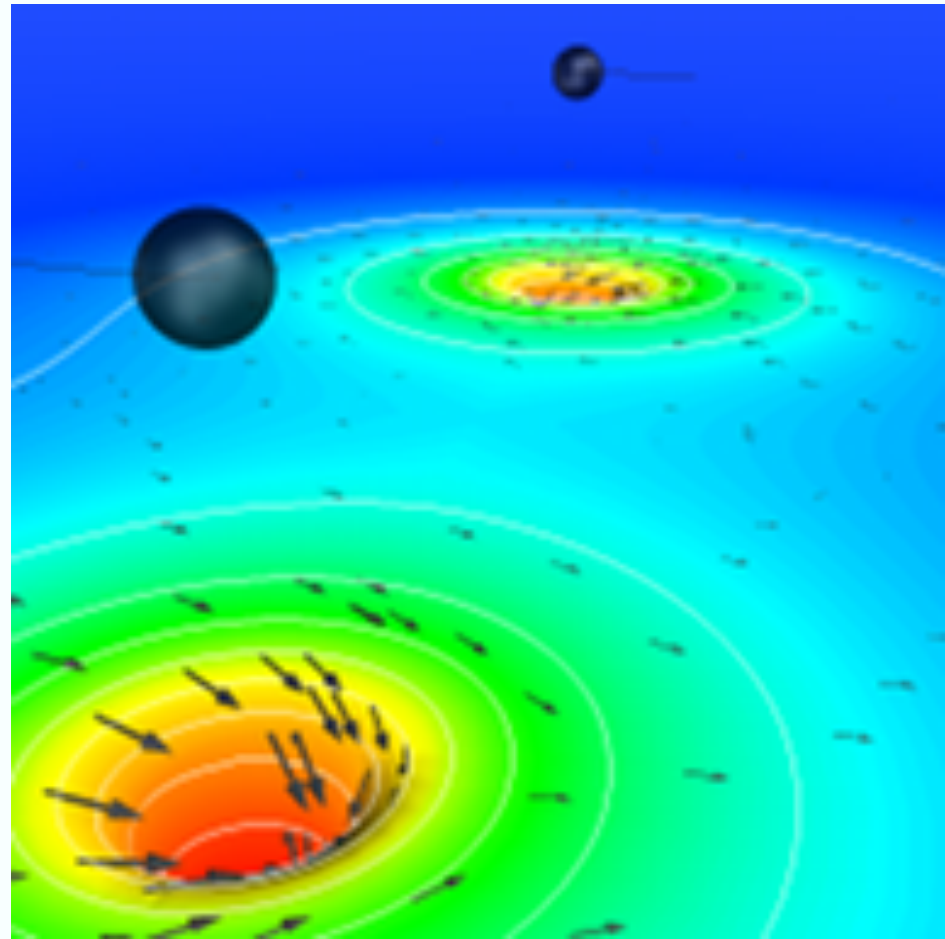
NR

PN

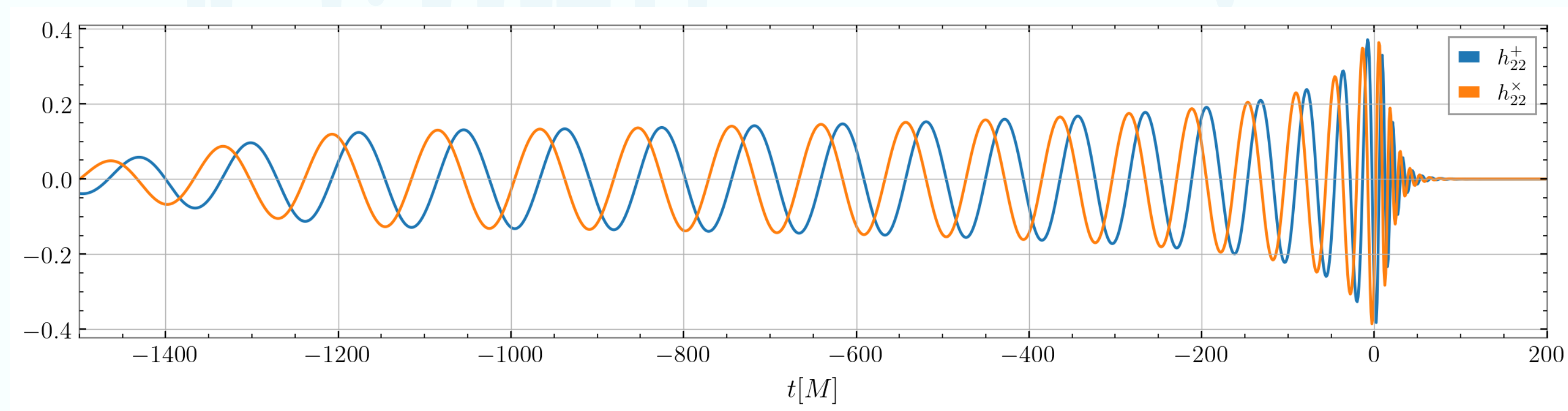
PM

EOB

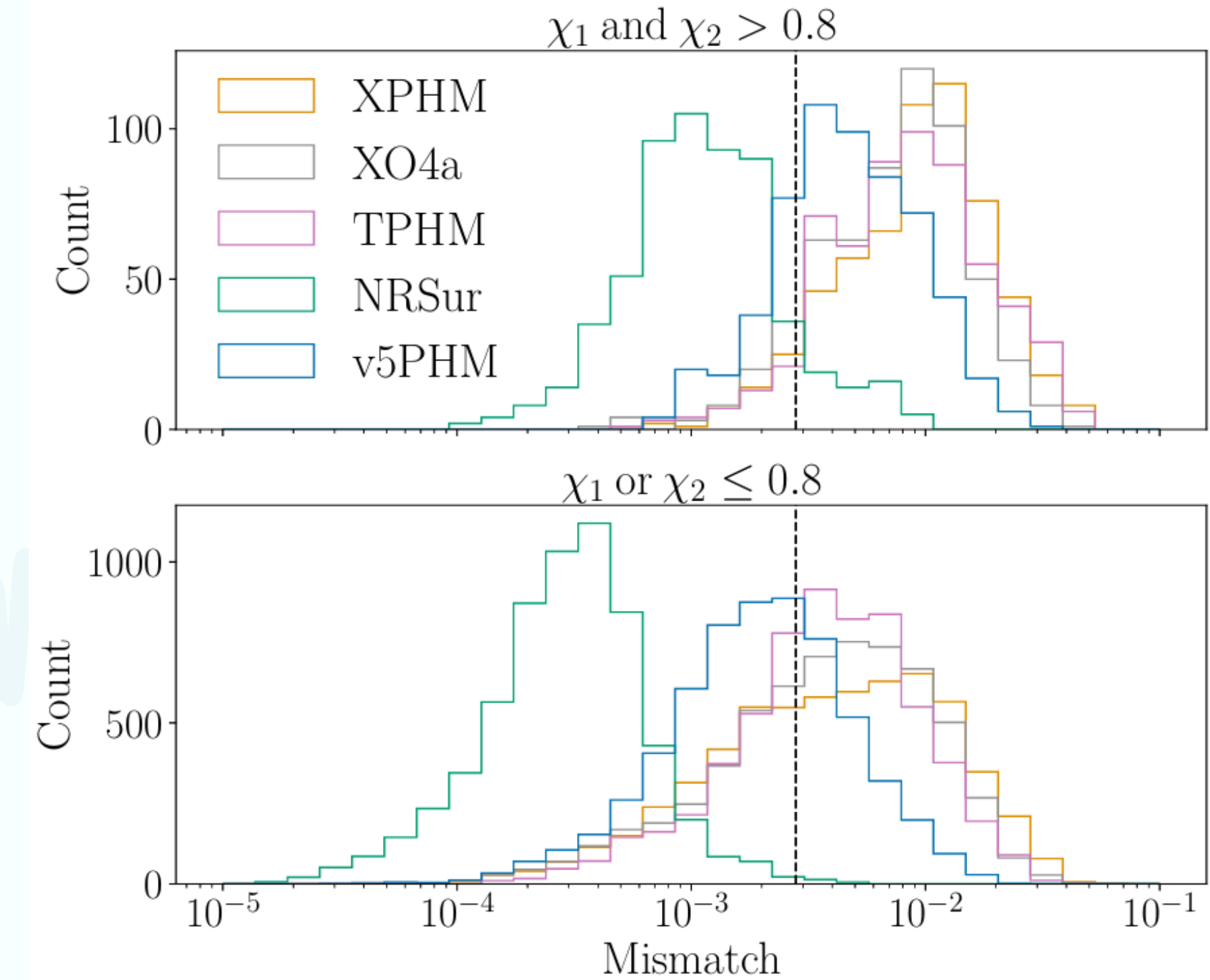
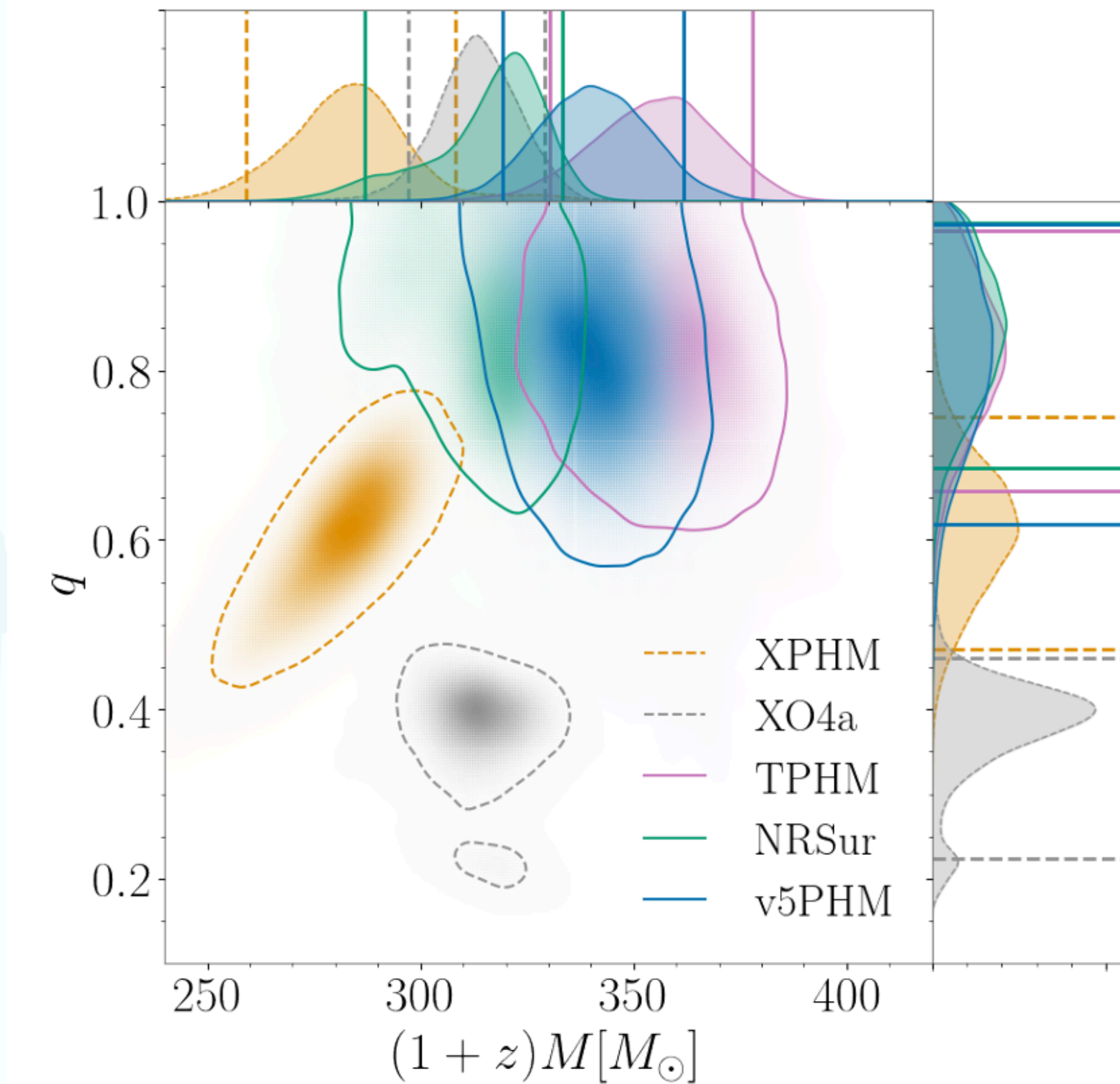
Numerical relativity



SXS/SpEC

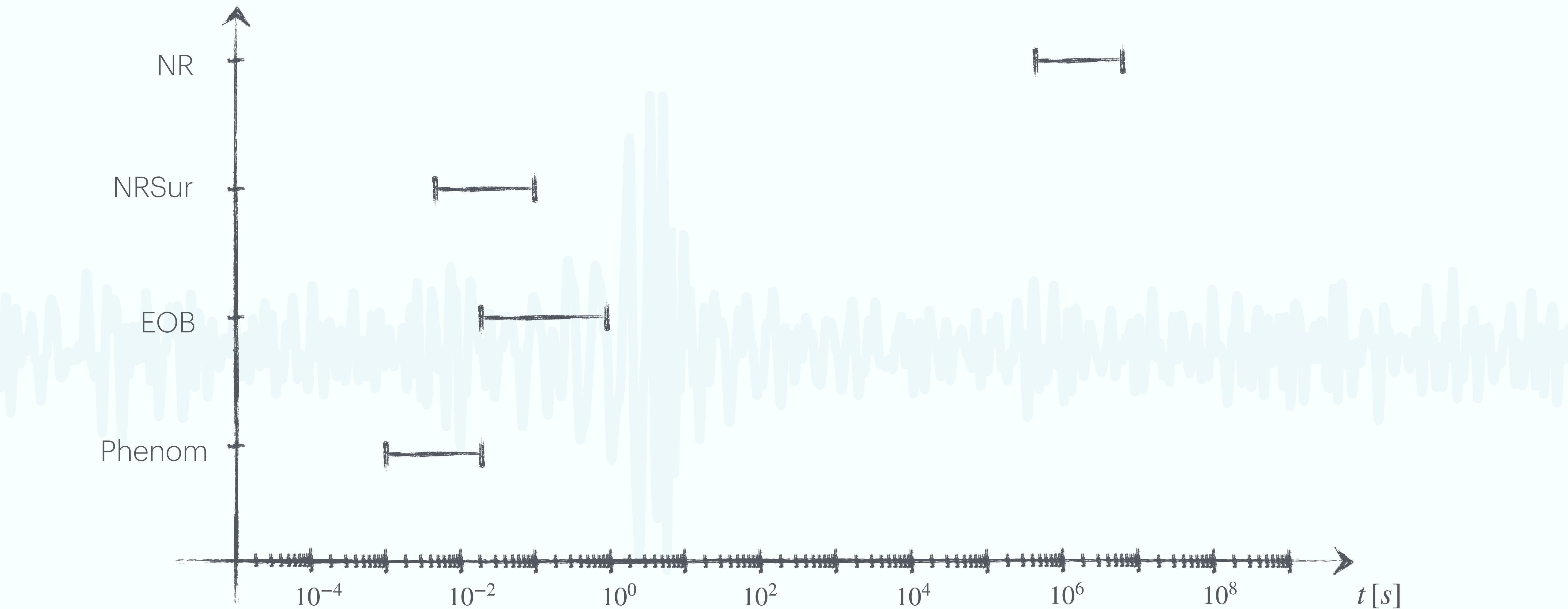


Waveform systematics



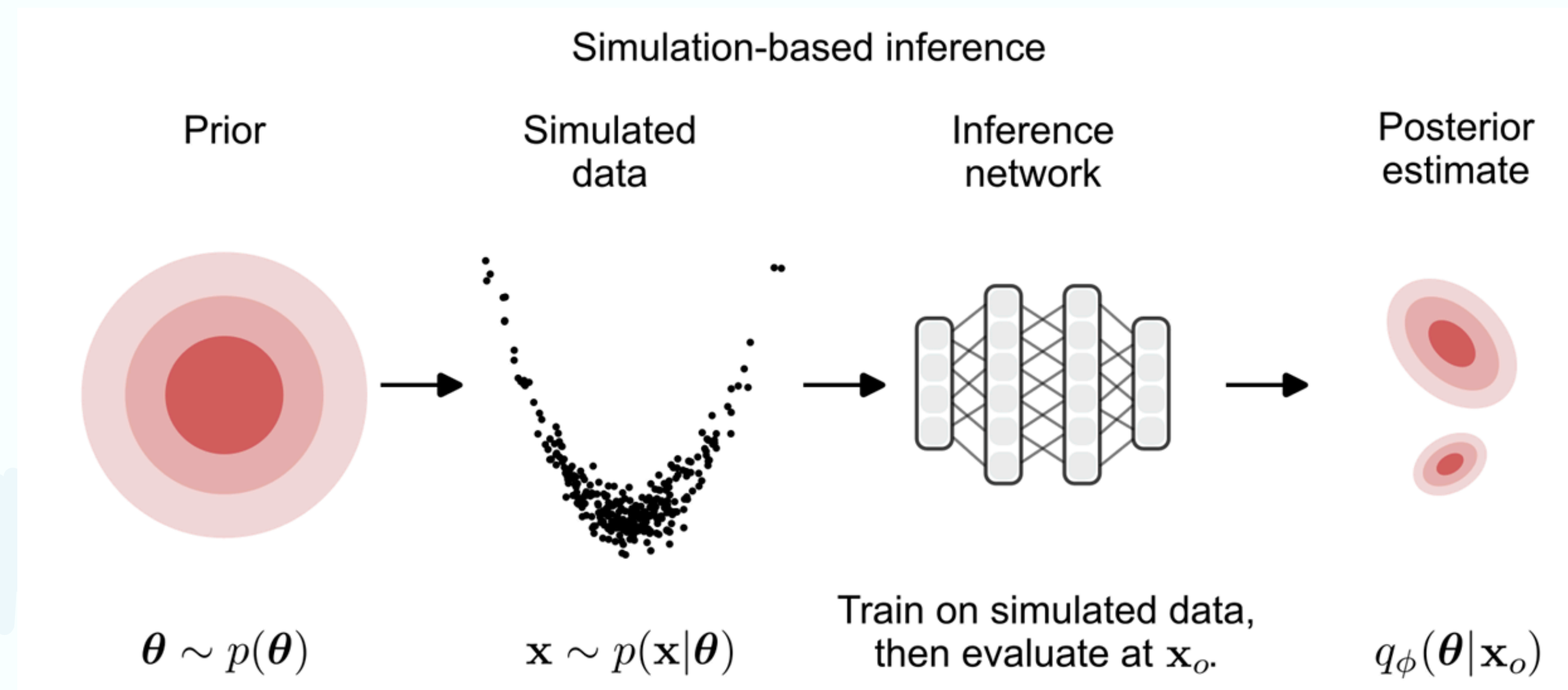
Abac+25

PE with NR?



Handmade,
probably inaccurate

SBI comes...

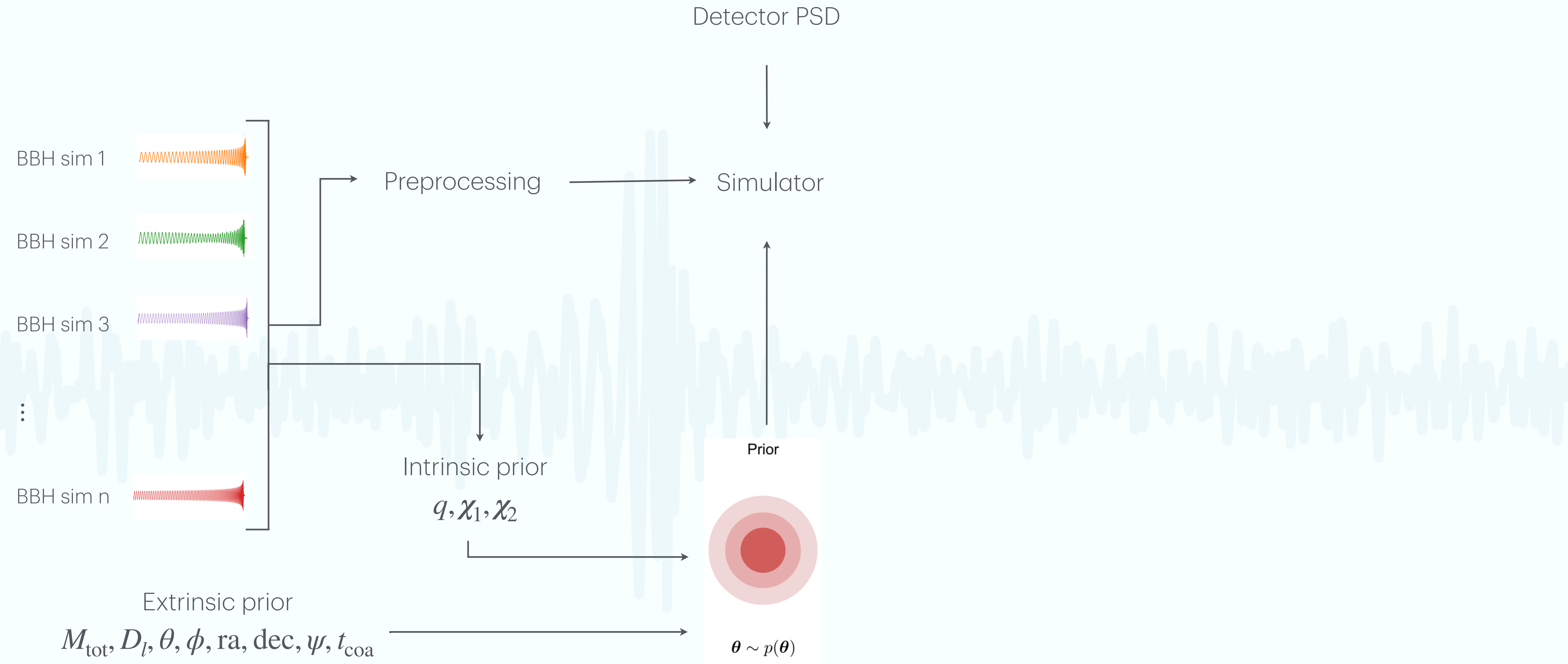


Deistler+25

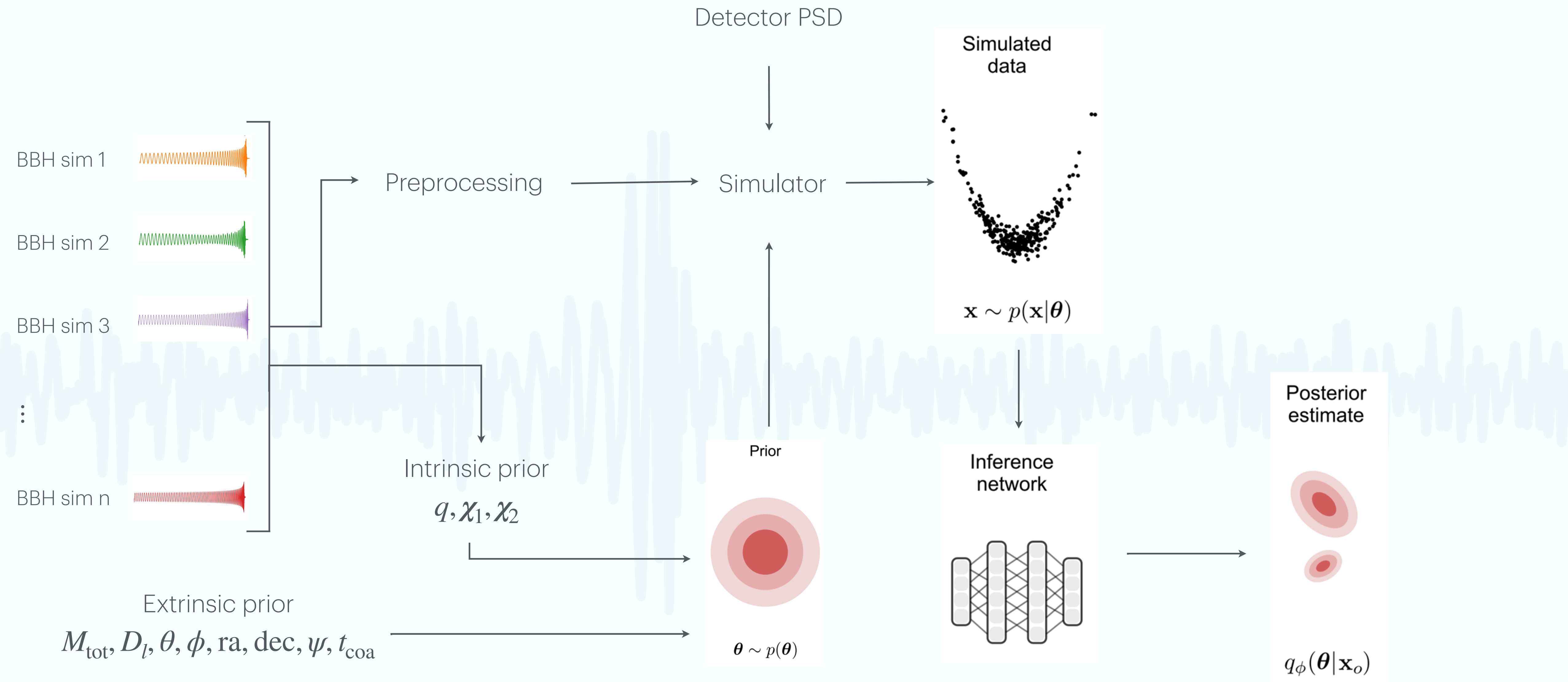
...with ideas...



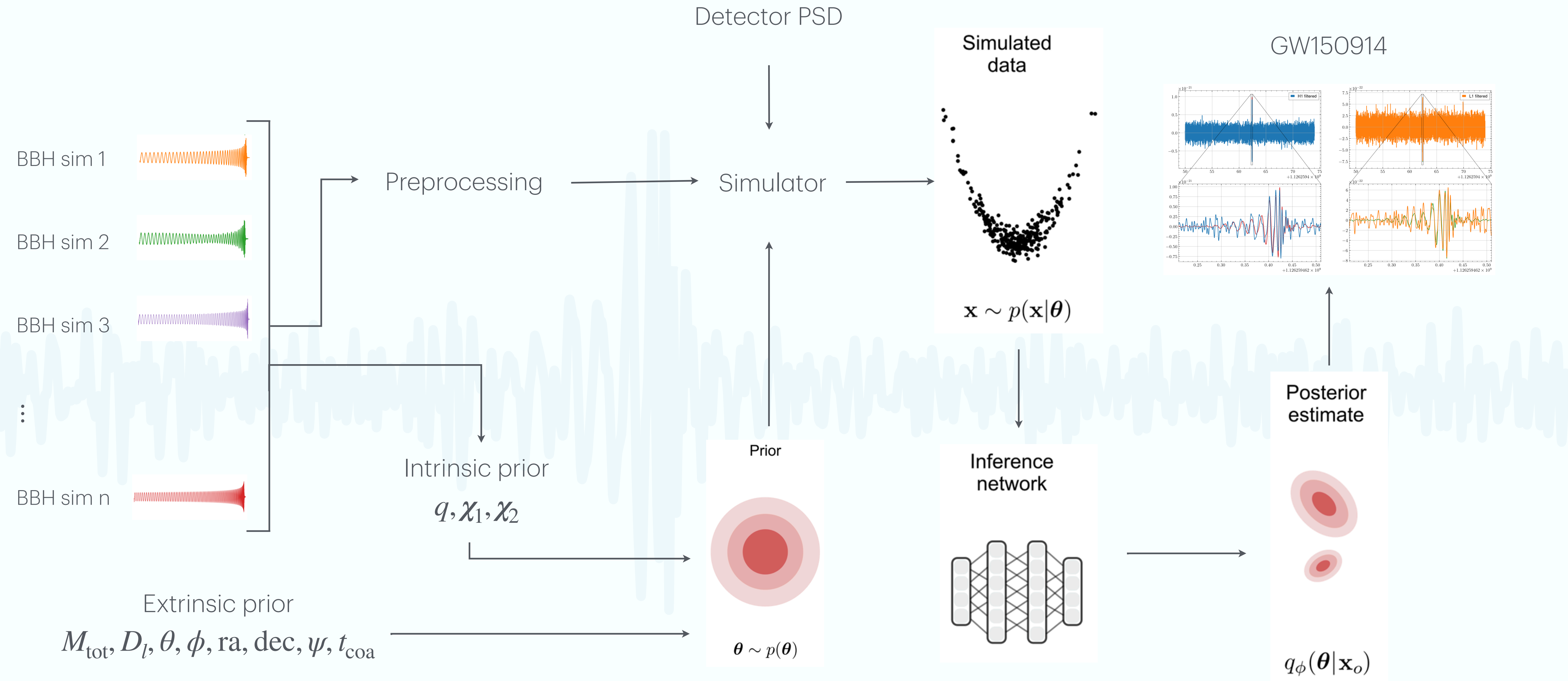
...with ideas...



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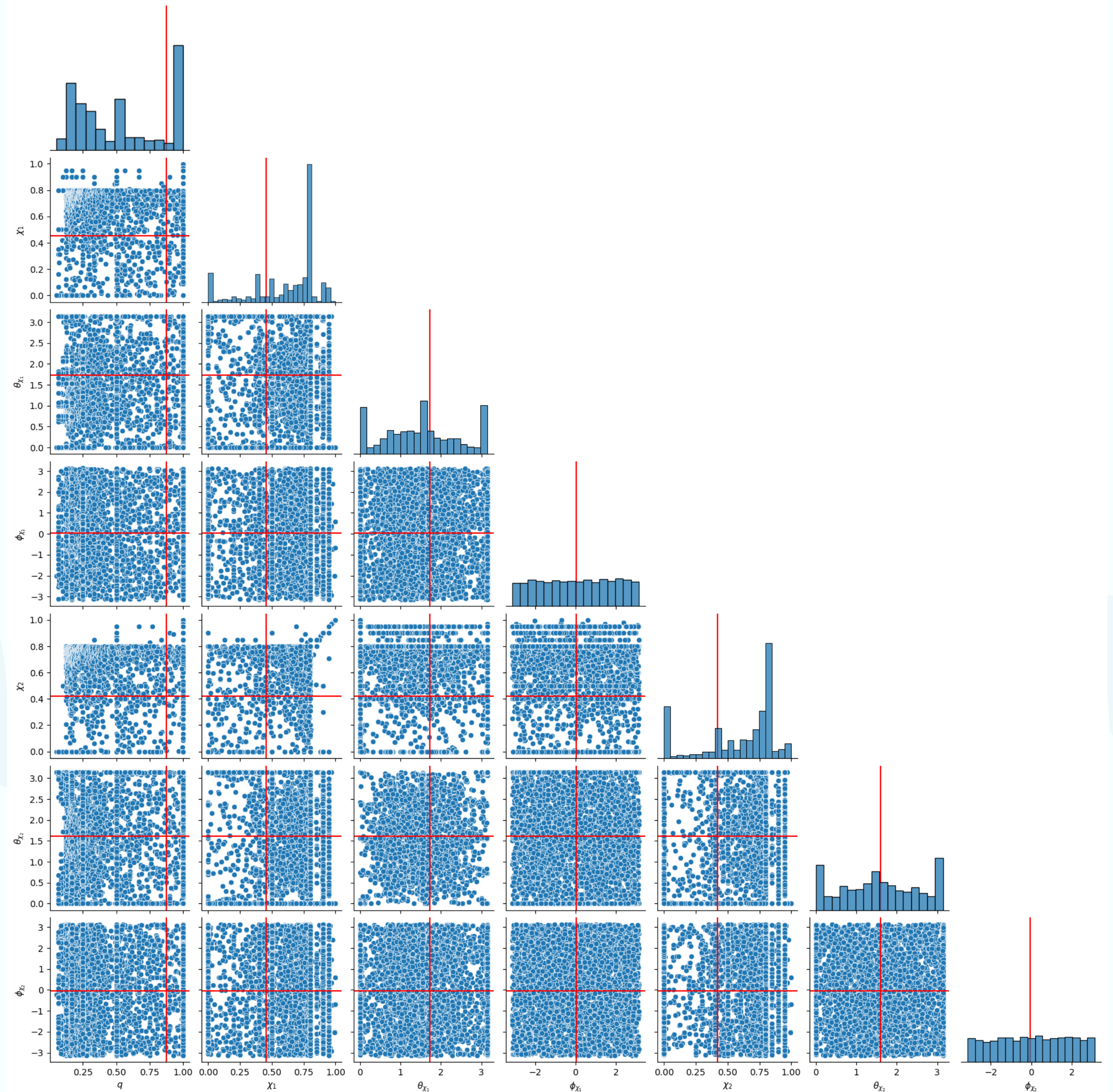


...with ideas...



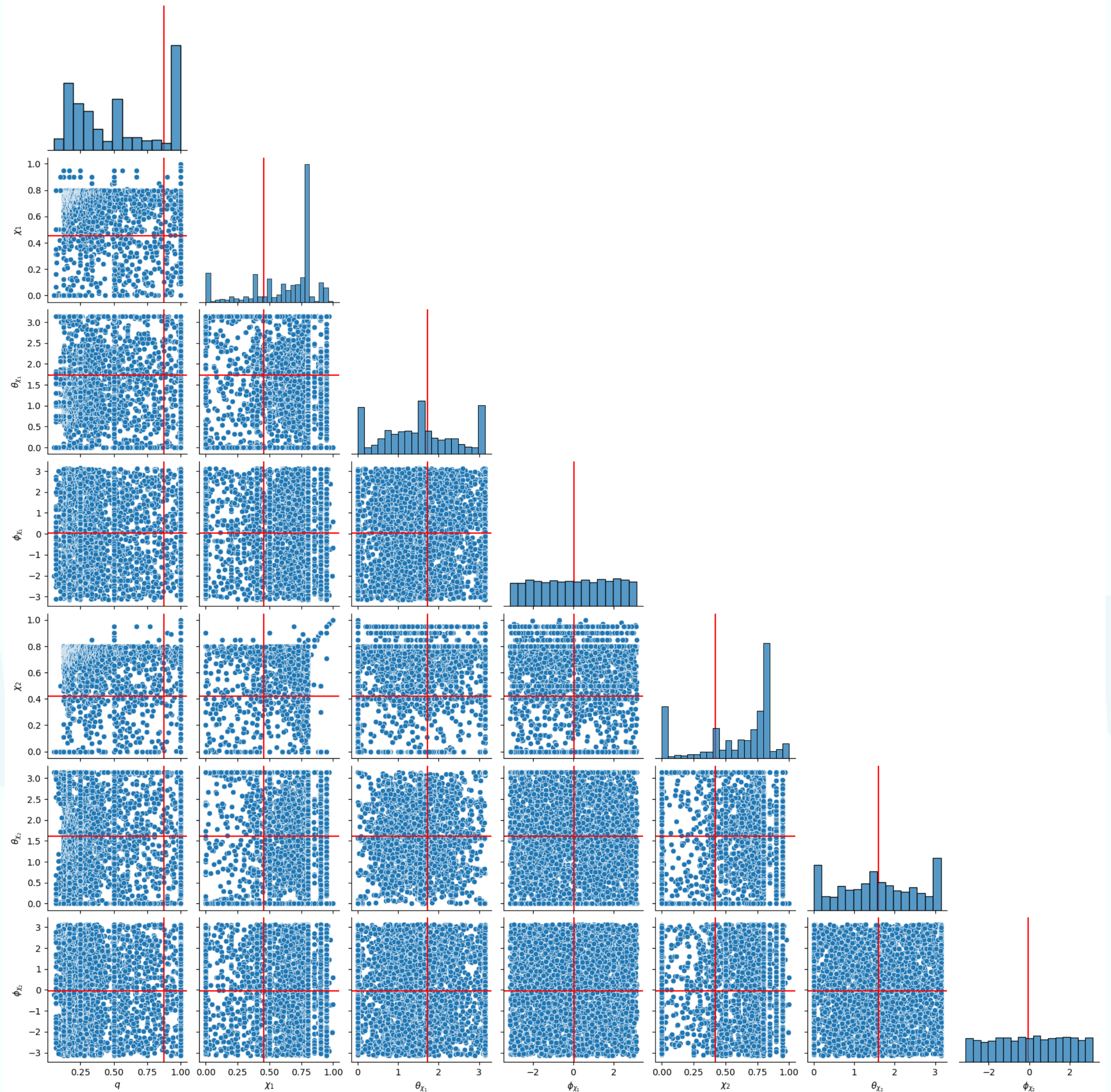
...and challenges

- Simulations are already done ✓✓✓
- “Only” ~4k simulations to train the network ✗✗
- Discrete coverage of intrinsic parameter space ✗
- Sample extrinsic parameters for free ✓
- Find the optimal architecture ?



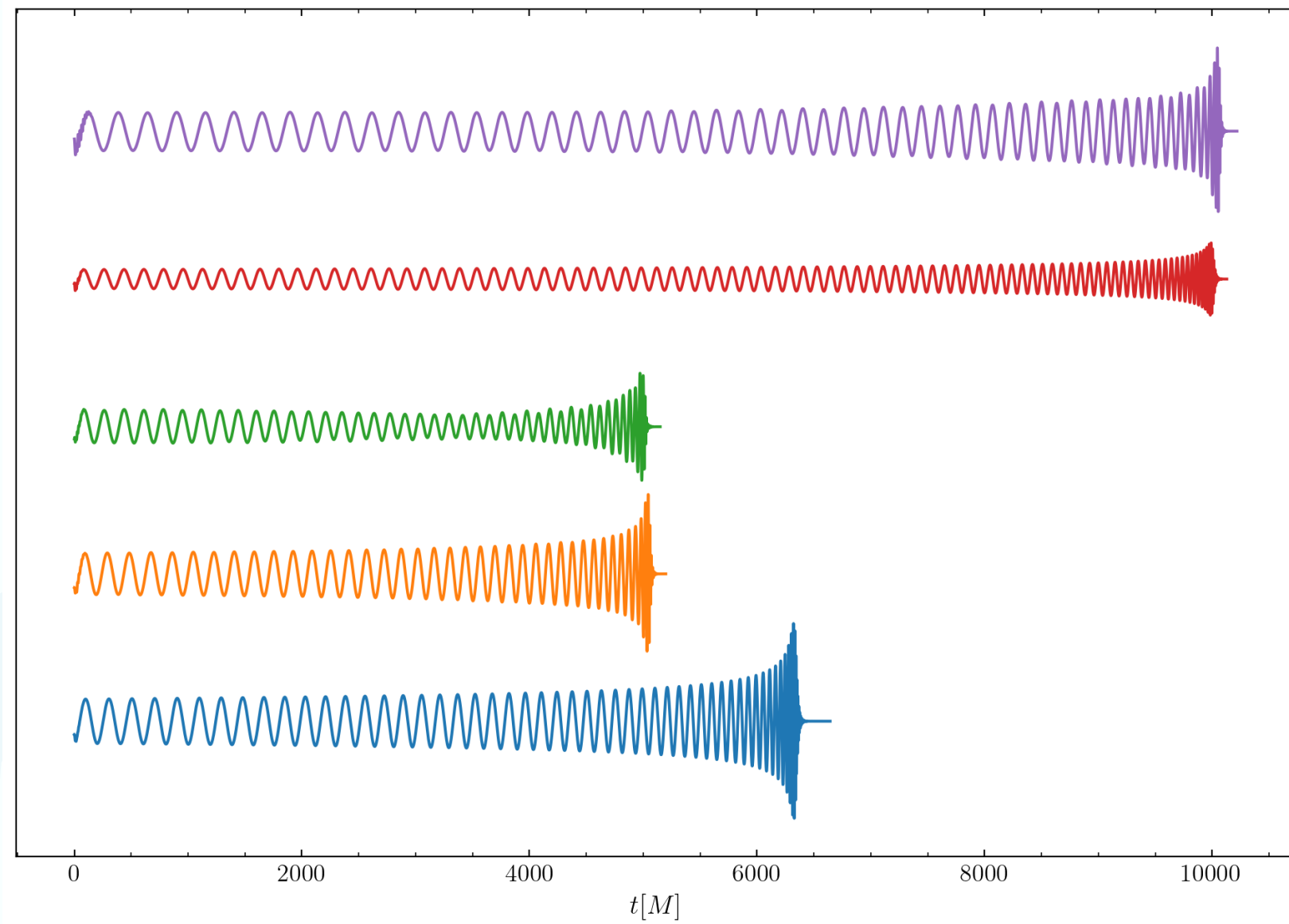
Waveforms database

- 3932 IMR simulations from SXS catalog
[Scheel+25](#), [Boyle+19](#)
- Quasi-circular BBH with $1250M$ -long
inspiral
- Preprocessing & standardization
- Frame rotation & spin evolution



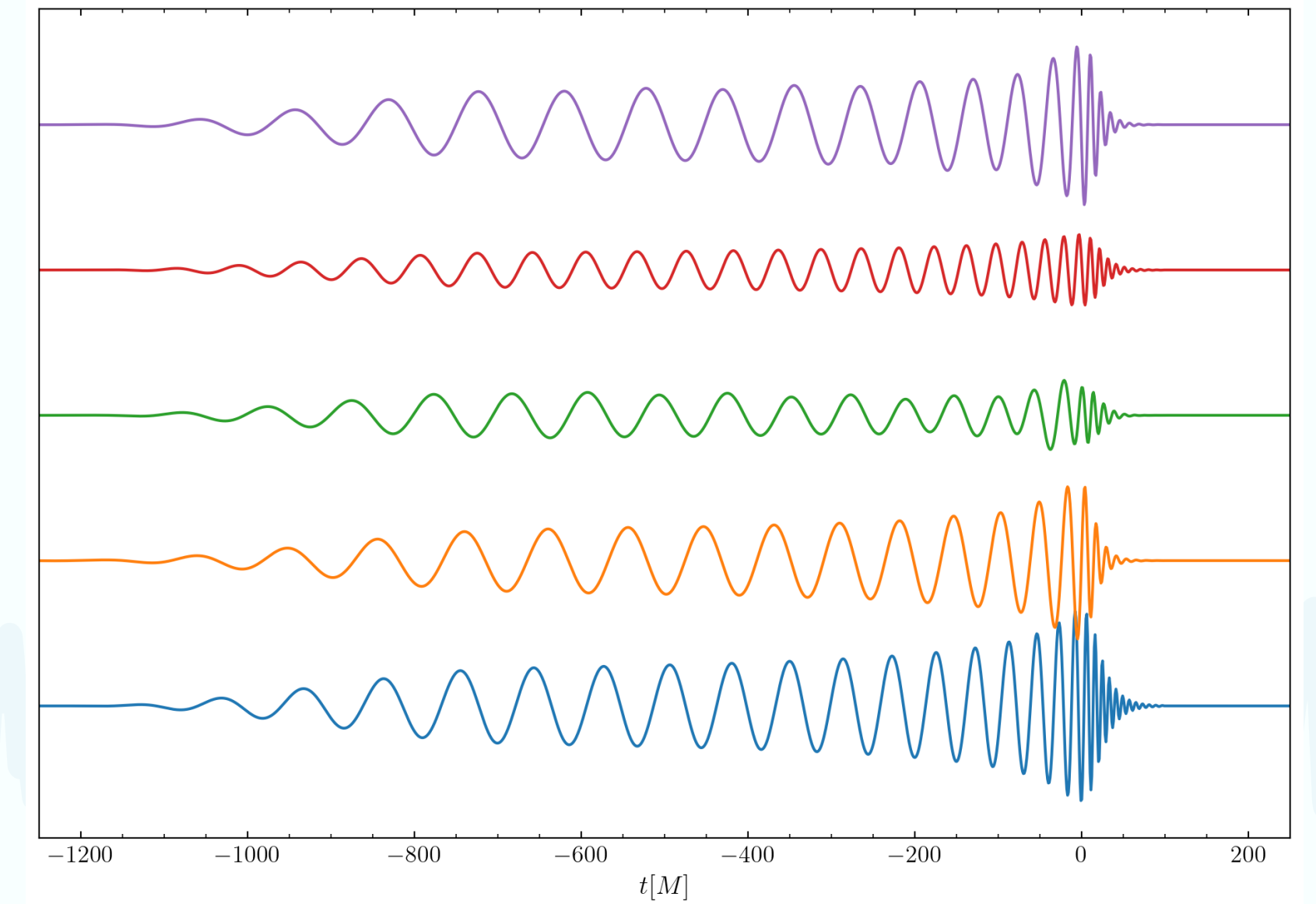
Waveforms database

$$l \leq 4$$



Preprocessing:

- Interpolation
- Tapering
- Padding
- (Zeroing memory)

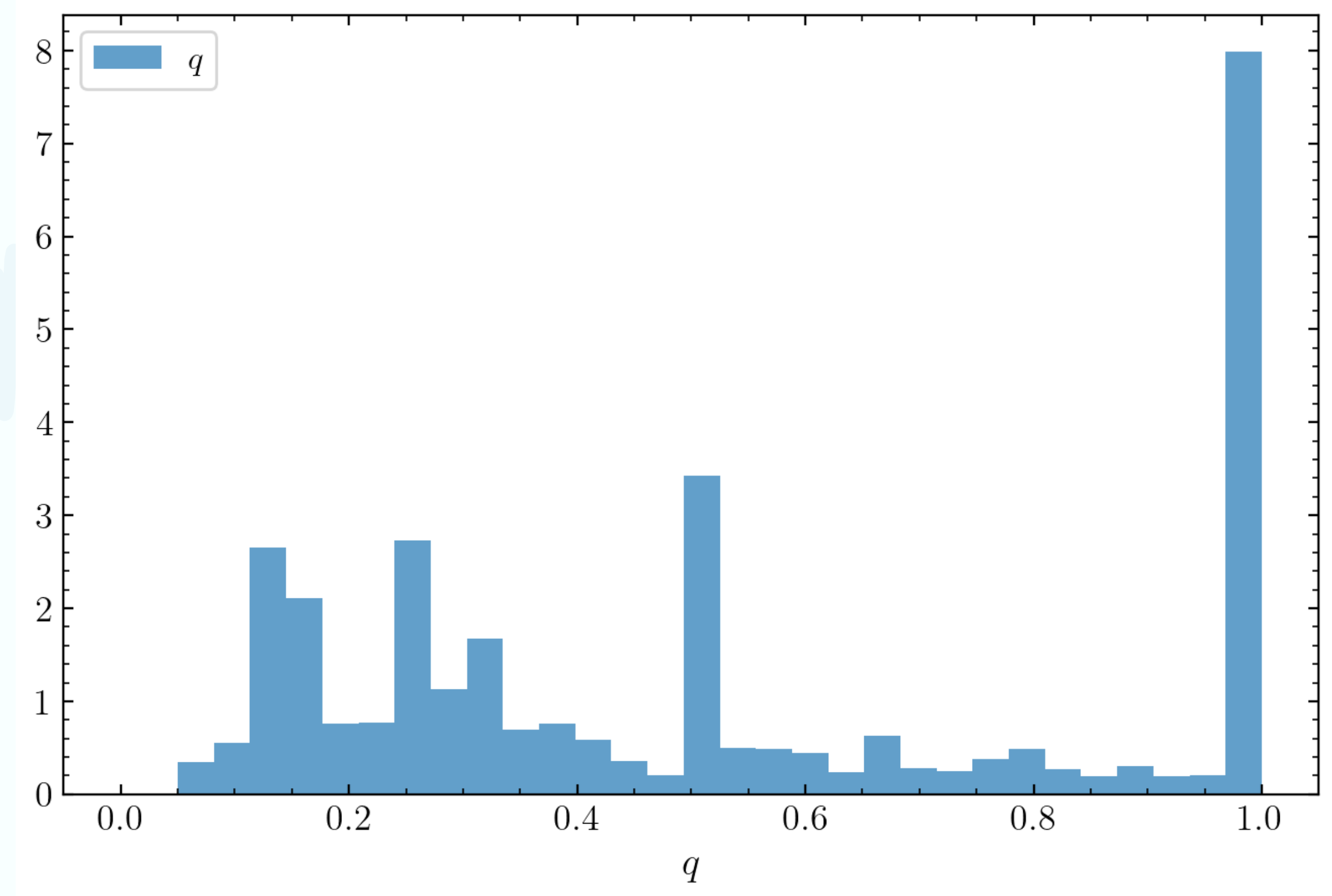


+

Frame rotation to account for spin definition

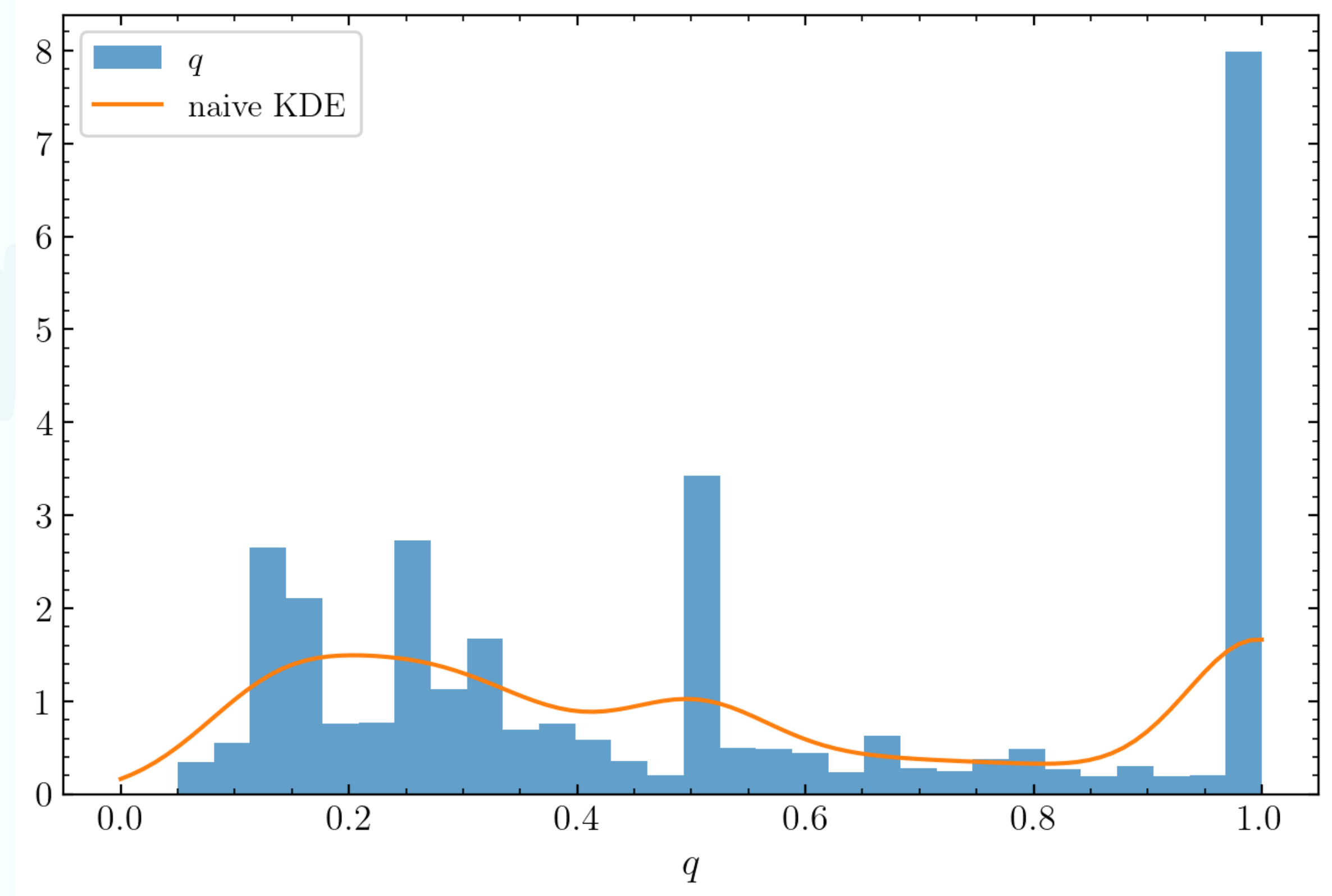
How to deal with a discrete prior

- $m_1, m_2 \sim U[10, 80] M_\odot \longrightarrow q, M_{tot}$



Intrinsic prior

- $m_1, m_2 \sim U[10, 80] M_\odot \longrightarrow q, M_{tot}$ **XX**
- KDE(θ^{intr})



Intrinsic prior

• $m_1, m_2 \sim U[10, 80] M_\odot \longrightarrow q, M_{tot}$ **XX**

• $\text{KDE}(\theta^{intr})$ **X**

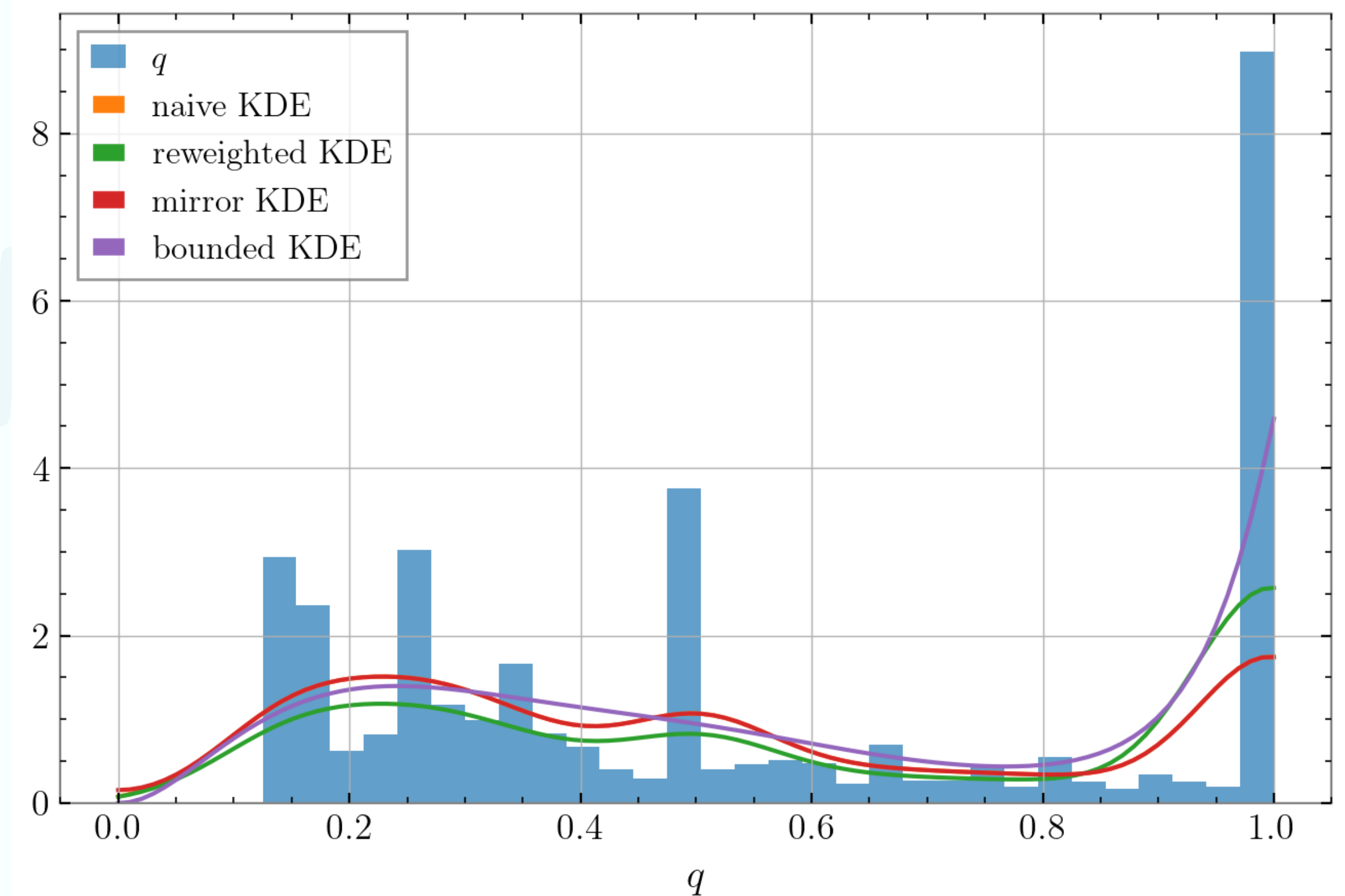
• Mirror $\text{KDE}(\theta^{intr})$ **✓** / **!**

• Reweighted $\text{KDE}(\theta^{intr})$ **!**

• Bounded $\text{KDE}(\theta^{intr})$ **✓**

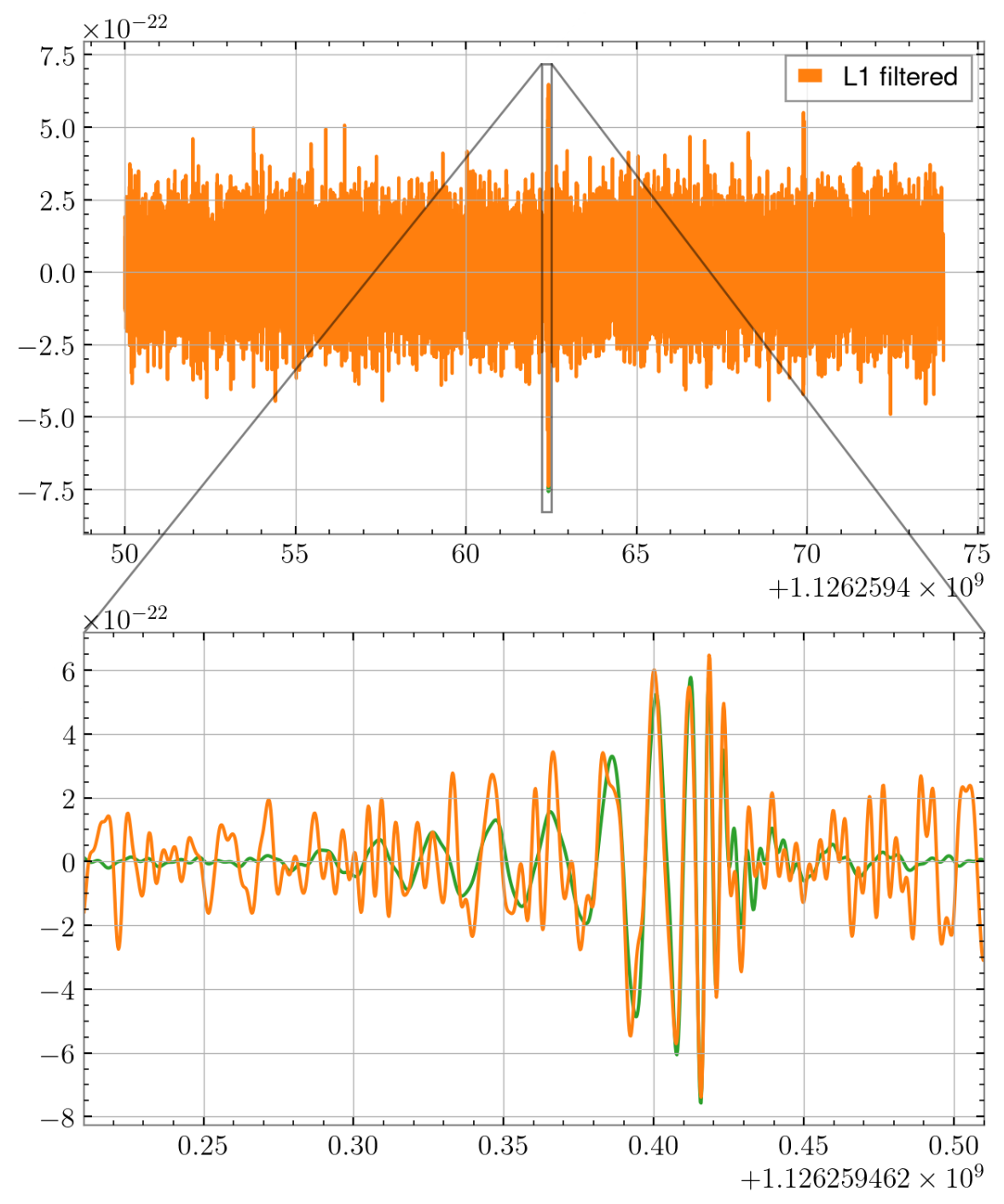
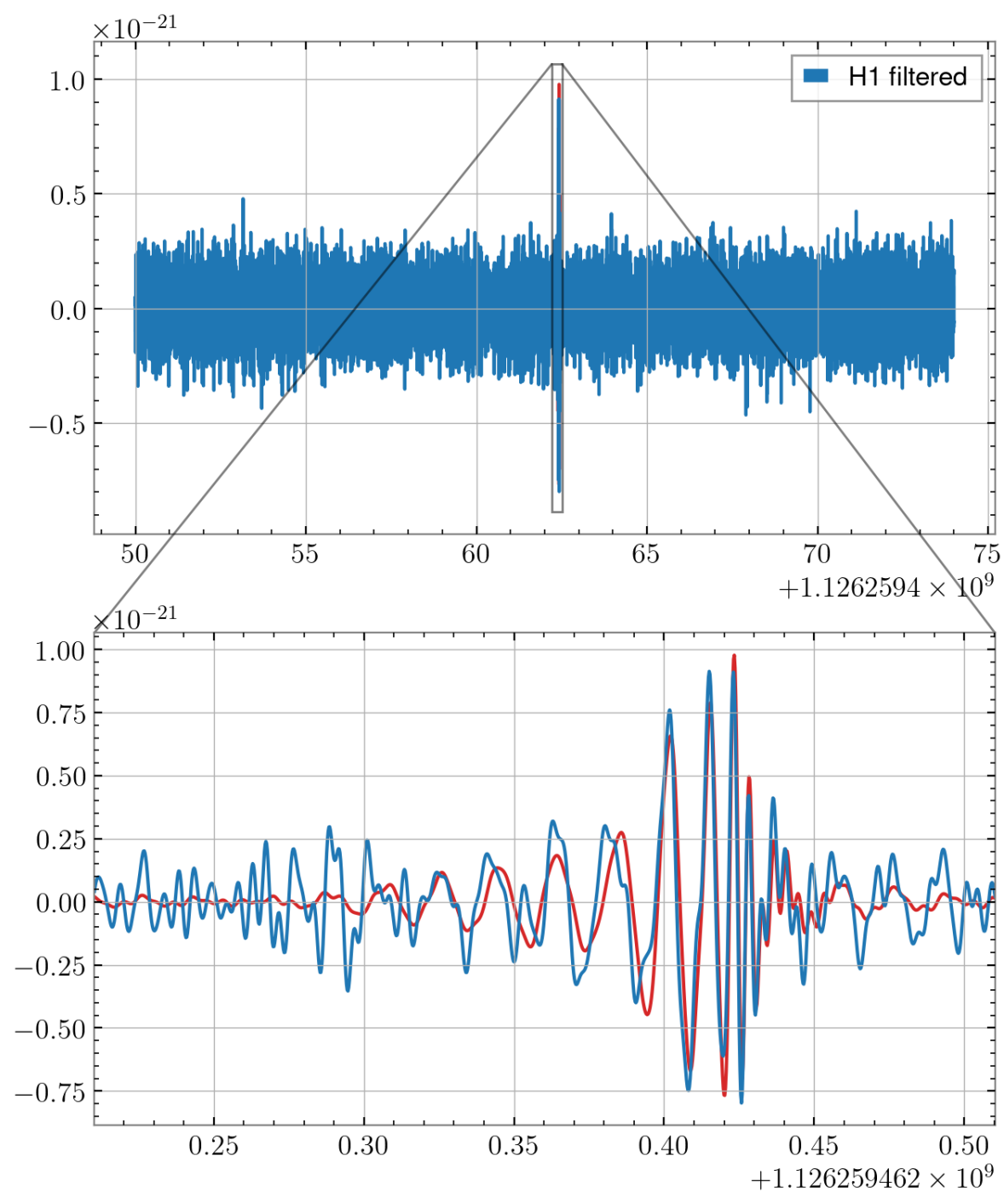
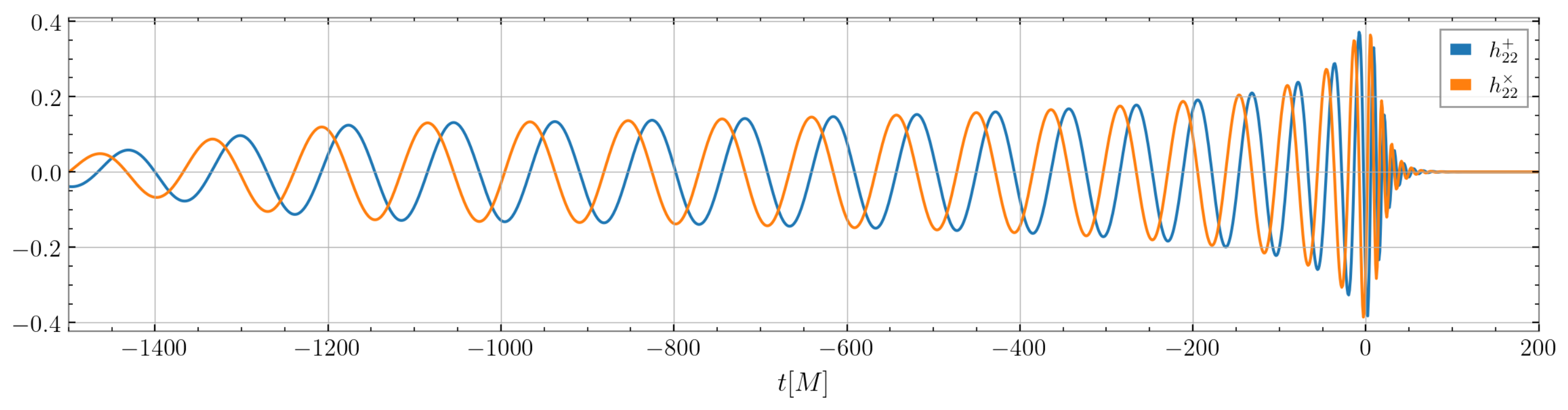
+

uninformative extrinsic priors

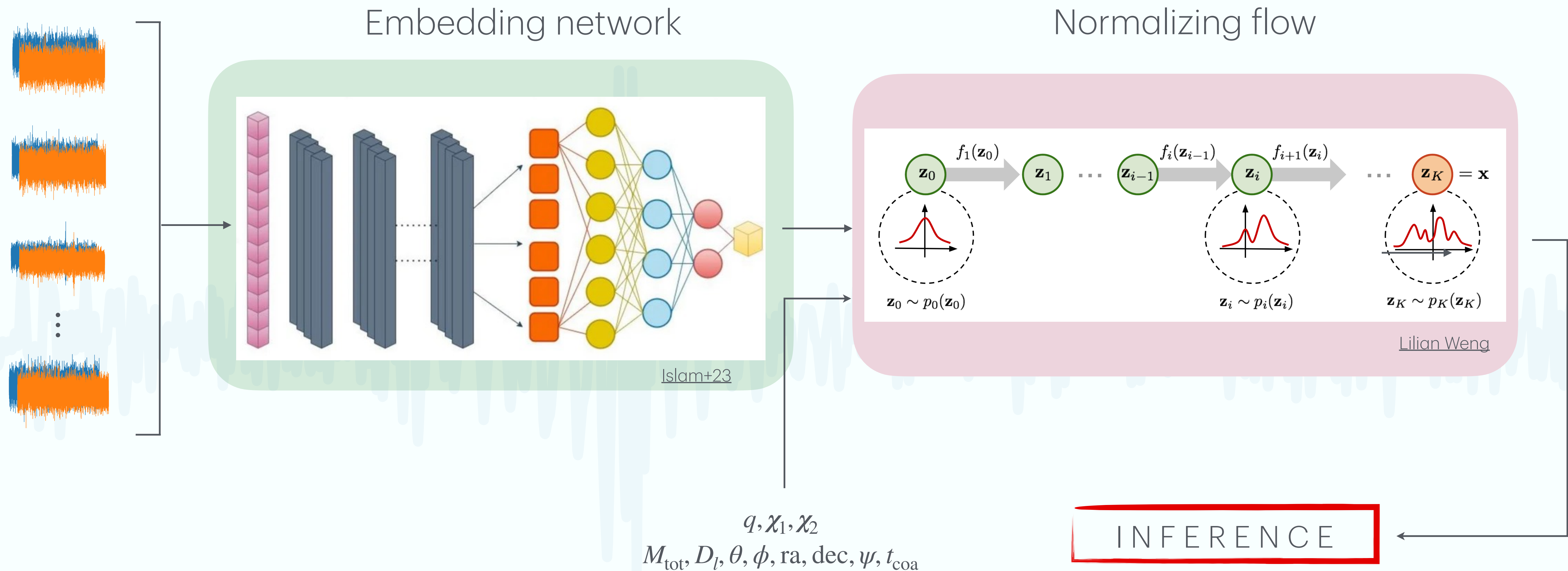


The beauty...

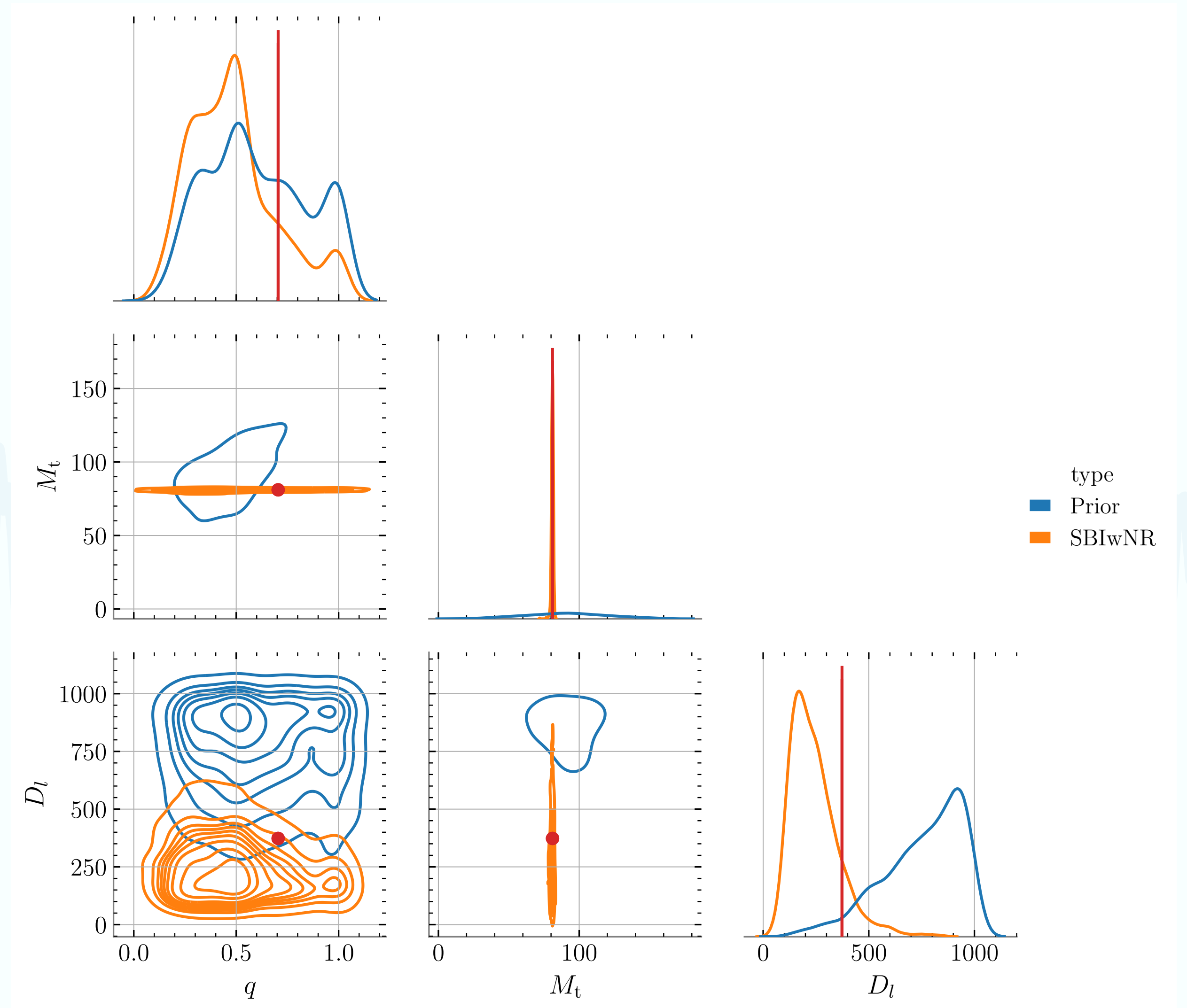
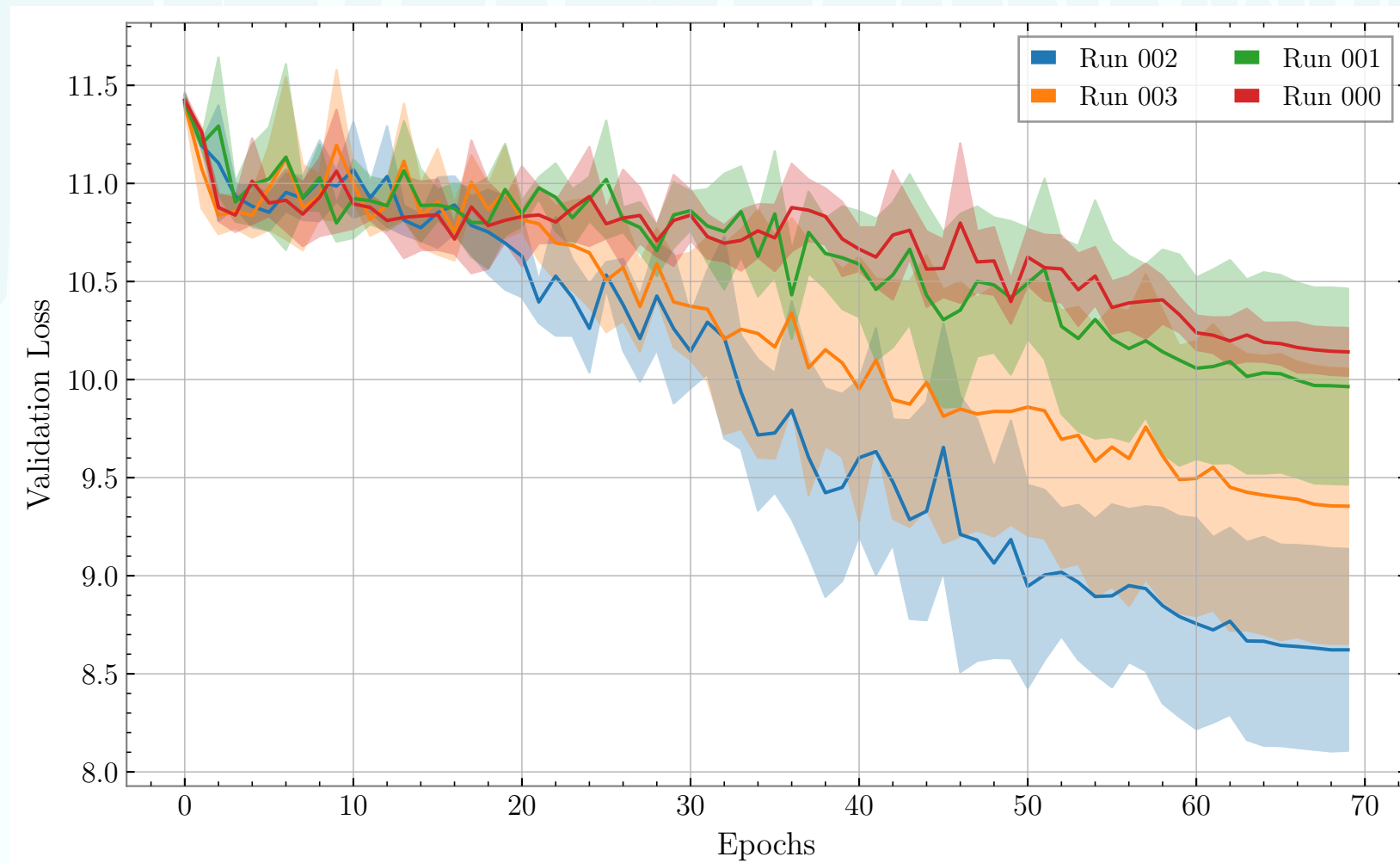
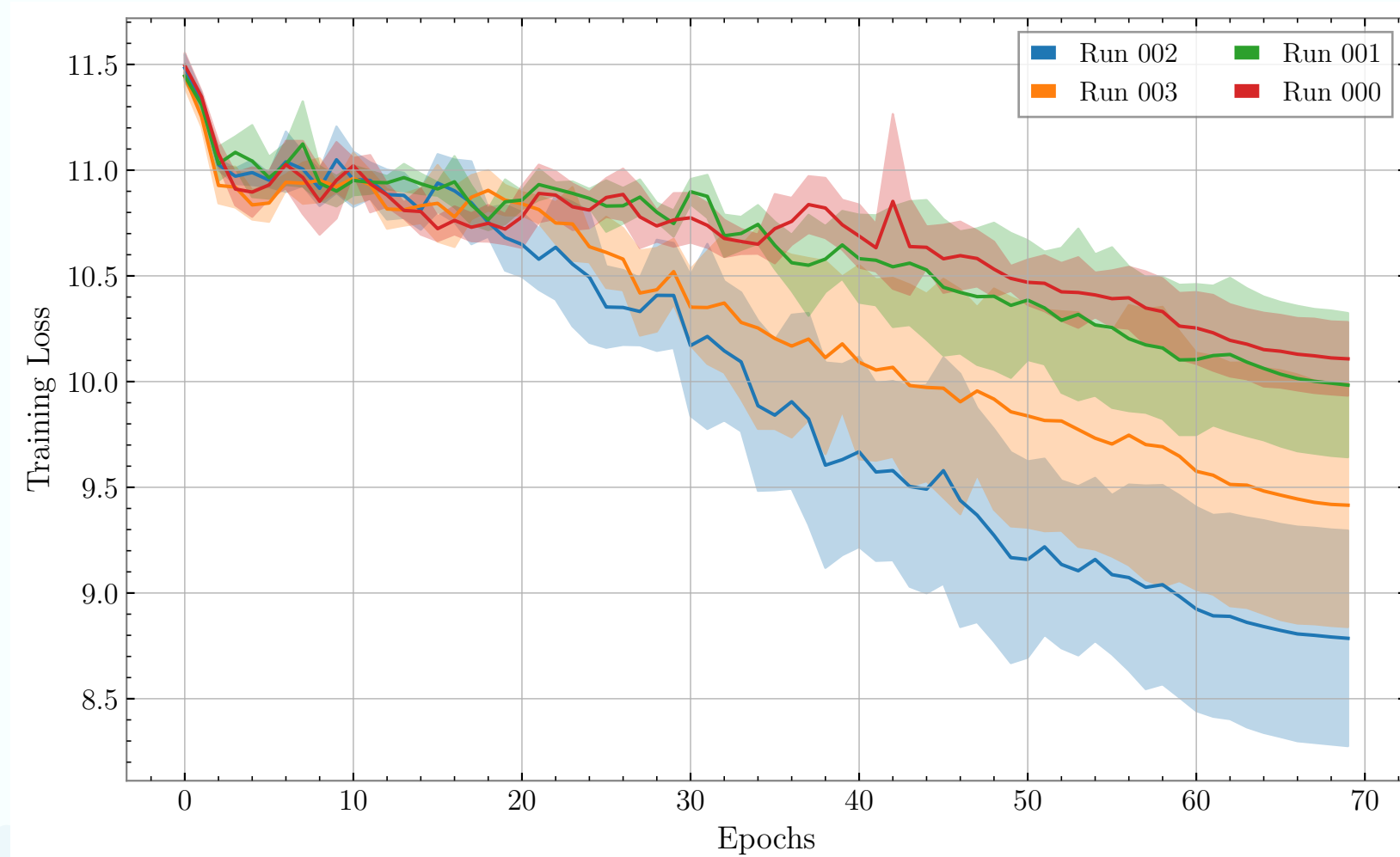
Not inference (yet)!



...and the Beast



(Very very) preliminary results



GW astronomy



without waveform “approximants”

Thanks for your attention!